# Grounded: Discrimination and Lifecycle Outcomes of America's First Black High Flyers*

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**Abstract**

We study the economic cost of racial discrimination using a natural experiment from World War II. In 1941, the US Army Air Forces began training African-American combat pilots at Tuskegee, Alabama. These men passed the same cognitive tests, physical screenings, and flight evaluations as white cadets. After the war, these highly selected Black pilots returned to counties that differed sharply in racial discrimination. We track their outcomes over the life cycle using census records, city directories, public address records, and death records. Higher racial discrimination was associated with Black pilots working in lower-paying occupations, with no corresponding penalty for white pilots. Differences remain visible in later occupations, housing values, and neighborhood income at death. To address sorting into locations, we use an instrument based on quasi-random wartime squadron peer origins, which affected initial postwar settlement.

**Keywords:** Discrimination; Tuskegee Airmen; Life-Cycle Outcomes; High-Skill Workers

**JEL Classifications:** J15, J71, J24, N32, R23

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> The Tuskegee Airmen were probably the most talented group of African-American men ever brought together in one place.
>
> — *The New York Times,* May 7, 1995

# 1. Introduction

Large Black–white gaps in earnings and wealth persist in the US (Bayer and Charles, 2018; Derenoncourt et al., 2024). Understanding the role of discrimination is difficult because it can affect both the acquisition of skills and the opportunities available to workers with those skills. Adjusting for pre-market differences in ability substantially reduces some estimates of racial earnings gaps (Neal and Johnson, 1996), but such differentials themselves reflect the impact of "systemic discrimination" (Bohren et al., 2025; Small and Pager, 2020). Skill and ability gaps are partly driven by schooling differences and differential exposure to unfavorable environments including the long-run aftereffects of Jim Crow laws (Chetty et al., 2020; Collins and Margo, 2006; Althoff and Szerman, 2026).

To isolate discrimination after labor-market entry, an ideal experiment would expose a cohort of men with similar pre-market characteristics to varying levels of discrimination. Using a set of outcomes over the life cycle, one would then compare the discriminated group's outcomes across environments, using the non-discriminated group as a benchmark. Such settings are rare.

In this paper, we exploit a natural experiment from WWII that comes close to the ideal experiment. In 1941, for the first time in history, the US Army Air Forces (AAF) allowed African Americans to attend military pilot training at a segregated base near Tuskegee, Alabama (Haulman, 2015). Around 1,000 Black men, who became known as the Tuskegee Airmen, passed the same physical screenings, cognitive tests, and flight evaluations as whites.[1] The AAF washed out 40–50% of cadets (USAAF Assistant Chief of Air Staff, 1943; Pierce, 2013; Haulman, 2015). Civil rights leaders insisted that Black cadets meet the same standards as white cadets, fearing that lower requirements would reinforce racist claims that Black men lacked the ability to become military pilots. By the time they earned their

[1] See Section 2.1 for more details.

wings, the selection process had produced a group of Black pilots with high and comparable skills. After the war, most pilots were discharged and returned to different parts of the country: some to the Jim Crow South, others to the relatively more open North. A group of Black men with *similar demonstrated skills* thus entered civilian life in places with *varying levels of discrimination*.

To fix ideas, consider the contrasting fates of two Tuskegee pilots: Terry Bailey and Lee Archer. Both were highly skilled. Bailey returned to Richmond, Virginia, where his parents lived—a city with a long history of discrimination against African Americans. He became a postal courier. Archer, by contrast, returned to Westchester, New York. He eventually became the first African-American vice president of a major US corporation. These divergent outcomes are not simply due to Westchester being richer overall. For white pilots, postwar outcomes could be remarkably similar across locations: John Whiteman—like Archer, from Westchester, NY—owned a billboard company. His fellow Eighth Air Force pilot Noah Tipton—like Bailey, from Richmond, VA—did not fare worse: He owned three companies, including a small commercial airline.[2]

Our empirical analysis generalizes this set of comparisons with a linked life cycle dataset. We follow Black and white WWII pilots from 1950 outcomes through mid-career occupations to late-life residential and neighborhood outcomes.

Postwar outcomes of Tuskegee Airmen varied with the level of racial discrimination in the counties where they settled. Outcomes for white pilots serve as a benchmark. We construct a composite index of pre-1945 racial discrimination at the county level to capture persistent differences, drawing on multiple historical indicators of racial hostility and segregation (Qian and Tabellini, 2024). There is evidence of talent misallocation for Tuskegee Airmen from 1950 until death. A one standard deviation (SD) increase in county-level discrimination (roughly half the difference between Westchester and Richmond) is associated with a 0.9 SD decline in occupational status for Black pilots relative to white pilots. The corresponding difference in occupational median income is approximately $1,000 in 1950 dollars. This gap reflects occupational misallocation: in higher-discrimination counties, Black pilots worked in occupations below their skill level. In the 1990s, the gap in the national ranking of house prices is 5 percentiles. By the time they passed away, a one SD difference in discrimination still

[2]See Bailey's story here, Archer's here, Tipton's here, and Whiteman's here.

corresponds to 9 percentiles in the average income ranking of the death ZIP code.

Can these results be interpreted as causal? Pilot selection compressed skill differences, but it did not eliminate them entirely. If higher-ability Tuskegee Airmen disproportionately sorted into lower discrimination counties, our estimates would overstate the effect of discrimination. To identify causal effects, we use variation in squadron peers' birthplaces to instrument pilots' initial postwar discrimination exposure. The AAF assigned pilots to squadrons by race and pilot type. Who served with whom was mainly driven by wartime recruitment pressures and high washout rates. Conditional on these assignment factors, the home counties of a pilot's squadron peers are plausibly random. Wartime social ties shaped postwar settlement decisions.[3] By 1950, pilots were significantly more likely to have settled in counties of their squadron peers. We instrument 1950 residence-county discrimination with the weighted average discrimination level of a pilot's squadron peers' birth counties. The underlying assumption is that pilots whose social networks included low-discrimination locations found it easier to "move to tolerance." The resulting 2SLS estimates show consistently large and significant effects of residence-county discrimination on socioeconomic status in 1950.

Several theories predict that ability should cushion the effect of discrimination. Under statistical discrimination, employers fall back on race only because individual productivity is hard to observe (Phelps, 1972; Arrow, 1973). The most able, though, have more ways to signal what they can do—stronger credentials, higher test scores, visible records of achievement—so race should carry less weight for them, and the racial gap should be smallest at the top of the ability distribution (Arcidiacono et al., 2010; Bjerk, 2007). Our findings point the other way. Among 442,000 WWII enlistees with cognitive test scores, the Black-white gap in postwar outcomes widens with measured ability rather than shrinking (see Figure 1). Tuskegee Airmen—drawn from the very top of the distribution—remain far behind comparable white pilots. The pattern echoes Bertrand and Mullainathan (2004), who find that the callback advantage of white-sounding names grows with resume quality. The evidence suggests that higher ability is associated with higher costs of discrimination, not a cushion against it.

A second view emphasizes learning instead of signaling. Even if skilled Black

[3] Green (2025) shows similar geographic mobility patterns among WWII naval sailors.

workers meet discrimination at first, employer learning implies that the penalty should fade: as firms observe a worker's productivity, they should lean less on race, and genuinely able Black workers should catch up with their white peers (Farber and Gibbons, 1996; Altonji and Pierret, 2001). This is a prediction about dynamics, and testing it means following the same men over time. We do so with a life-cycle design, tracking a single cohort of highly skilled men to the end of life. The gap does not close. Racial disadvantage remains visible in later occupational and residential outcomes, with no sign of the convergence that employer learning models would predict.

The life-cycle design offers an additional advantage. Earlier work studied racial earnings gaps in the aggregate, pooling cohorts during periods of large structural change in the economy and in the returns to skill (Bayer and Charles, 2018; Heckman et al., 2003). Following a single cohort holds the composition of the treated group fixed and so avoids the effect of aggregate change between different cohorts. It also separates two questions that aggregate data blur: whether discrimination merely shaped where these men started—their initial job placement—or whether occupational misallocation remained visible across the rest of their lives.

# 2. Historical Background

## 2.1 The Tuskegee Airmen and Pilot Selection

During WWII, the Army Air Forces operated as a branch of the US Army (it became independent in 1947). The USAAF did not accept Black men as pilots before WWII. The Army remained racially segregated throughout the war: Black soldiers served in separate units and were often assigned to non-combat roles. Under pressure from civil rights organizations—most prominently the NAACP and the Black press—Congress passed legislation in 1939 designating funds to train Black pilots. In 1941, the USAAF established its Black pilot training program at segregated airfields near Tuskegee, Alabama (Haulman, 2015). Pilots who completed their training there became known as the "Tuskegee Airmen."

The selection process for military pilots during WWII was demanding. To be considered, an applicant needed at least two years of college attendance, a

threshold met by only about 5.5% of the US population in 1940. He also needed to pass a qualifying written examination testing practical judgment, reading comprehension, and mathematics (Du Bois, 1947; Flanagan, 1948, pp. 14–15). Candidates then faced a rigorous physical and psychiatric examination, requiring 20/20 uncorrected vision and strict standards for hearing, cardiovascular fitness, and neurological health (Du Bois, 1947, p. 2). Applicants who cleared these hurdles spent two full days at a classification center taking written exams on spatial orientation, mathematics, and mechanical reasoning, as well as apparatus tests of coordination and reaction time. Performance was summarized in composite aptitude scores, known as "stanines," that determined eligibility for pilot training (Flanagan, 1948; Du Bois, 1947, pp. 63–67). Of the more than 600,000 men who reached this stage during the war, over 150,000 were screened out before any flight training began (Du Bois, 1947, pp. 68–69). Those admitted still had to complete preflight ground school and primary, basic, and advanced flight training, where many of the cadets washed out (Pierce, 2013, p. xii). By the time a man earned his wings, the process had selected for high cognitive, physical, psychomotor, and sensory skills, as well as perseverance (National Center for O*NET Development, 2026; Duckworth, 2016). Fewer than 200,000 new pilots entered the USAAF during WWII.

Tuskegee Airmen were held to the same training and selection standards as white pilots. Civil rights leaders and the Black press demanded high standards at Tuskegee, rejecting any concessions that might reinforce racist claims that African Americans could not master complex technology (Scott and Womack, 1998). A wartime internal review stated plainly that "the requirements for Negroes were the same as for white applicants, and the procedure for examination was the same" (USAAF Assistant Chief of Air Staff, 1943, p. 21). Daniel Haulman likewise notes that Colonel Noel F. Parrish, the base commander at Tuskegee for most of the war, "insisted that the training of Black pilots be of the same quality as the training of white pilots at other bases." Only about half of Black flight cadets completed training. Those who graduated "were as fully qualified as any white pilots going through the same sequence of training at other bases."

In total, approximately 1,000 African-American men graduated from the Tuskegee program across 72 classes beginning in 1942. The rigor and uniformity of this selection process produced a group of Black pilots with unusually high and broadly

comparable skills. This makes the Tuskegee Airmen well suited to studying how postwar environments affected the returns to ability.

## 2.2 Postwar Discrimination and the White Pilot Benchmark

Racial discrimination varied substantially across US counties after WWII. In the South, Jim Crow laws had long enforced segregation in education, employment, housing, and public accommodations. In the North, discrimination was often less formal but still pervasive, operating through restrictive housing covenants, discriminatory hiring, and unequal access to public services. Some counties had active Ku Klux Klan chapters and long histories of racial violence; others had lower levels of residential segregation and smaller Black-white gaps in schooling. These local differences were highly persistent, so counties that were more discriminatory before the war tended to remain more discriminatory afterward (Acharya et al., 2016).

For Tuskegee Airmen, this geographic variation affected how fully their skills could be translated into civilian success. In higher-discrimination counties, barriers to skilled employment and home ownership were more severe; in lower-discrimination counties, those barriers were weaker (Fishback et al., 2024). As a result, men who entered civilian life with similar training and demonstrated ability could nevertheless experience very different occupational and economic trajectories depending on where they lived.

This geographic variation should have mattered much less for white pilots. Like the Tuskegee Airmen, roughly 193,000 white pilots passed through rigorous selection and training (USAAF Office of Statistical Control, 1945, Table 47), but they did not face comparable racial barriers in postwar civilian life. County-level discrimination may still correlate with worse outcomes for white pilots because more discriminatory counties also differed along other economic dimensions. The relevant comparison is therefore whether Black pilots' outcomes deteriorated relative to those of white pilots as local discrimination increased. White pilots provide a benchmark for distinguishing racial disadvantage from local economic conditions that affected both groups similarly.

# 3. Data

## 3.1 Pilot Dataset

We construct a baseline sample of Black and white WWII pilots with socioeconomic outcomes over the life cycle. Our baseline pilot dataset contains 4,327 pilots: 750 Black and 3,577 white pilots. All pilots in this baseline sample survived the war and were born between 1911 and 1925. This baseline sample is the starting point for all subsequent linkage to postwar outcome data.

We begin by assembling pilot samples, and then link them to the 1950 Census, City Directories, Index to Public Records, and death records. Together, these sources allow us to measure pilots' locations, occupations, 1950 income, housing values, and neighborhood quality from the immediate postwar period until death.

We construct a sample of Black pilots starting from a complete roster of all 1,009 Tuskegee Airmen compiled by the educational nonprofit CAF RISE ABOVE. The roster reports each pilot's name, hometown in military records, graduation year from Tuskegee, rank at graduation, and pilot type. Using genealogical searches on Ancestry, we recover birth year and birthplace for 903 of these pilots. Our linkage rate of approximately 90% is substantially higher than rates typically achieved using standard automated record linking procedures (Abramitzky et al., 2021). Table B.1 compares pilots matched and unmatched to Ancestry. The two groups are similar in graduation year, the share graduating as second lieutenants, and the share with hometowns in Southern states, although the matched sample contains a higher share of fighter pilots. Among the 903 linked pilots, we exclude 103 pilots who died by the end of 1945, since they cannot be linked to postwar records, and an additional 50 pilots born before 1911 or after 1925 to align the age distribution with that of the white comparison group.[4] The resulting Black pilot sample contains 750 men.

We construct a sample of white pilots using the same genealogical search

[4] Figure B.1 reports WWII survival rates: 89.3 percent of Black pilots and 78.0 percent of white pilots survived WWII. Figure B.2 and Figure B.3 compare years of schooling between pilots who survived WWII and pilots who died during the war. Among white pilots, schooling is very similar across the two groups; among Black pilots, those who died during WWII have slightly higher schooling. The surviving Black and white pilots also look broadly similar in schooling.

methodology. Because the AAF trained roughly 193,000 white pilots during WWII and no comprehensive roster is available, we focus on pilots who served in the Eighth Air Force. This choice is driven primarily by data availability: the Eighth Air Force has the most comprehensive large-scale collection of white pilot records we were able to assemble. It was also the largest AAF unit during the war and flew more sorties than all other air force units combined (Miller, 2001; USAAF Office of Statistical Control, 1945). We compile a list of 5,142 white pilots and recover birth year and birthplace for 4,641 of them (90%) through Ancestry. Section A.1 describes the underlying sources, and Figure B.4 shows that the matched sample contains a higher share of fighter pilots. Among the linked pilots, we then exclude 1,040 pilots who died by the end of 1945 and 24 born before 1911 or after 1925. The white pilot sample consists of 3,577 men.

*Linkage to Postwar Outcome Sources.* First, we manually link pilots to the 1950 Census.[5] We locate and transcribe legible census questionnaires for 2,563 pilots. These records provide county of residence, occupation, and income in 1950, and thus anchor our analysis at the start of pilots' civilian careers. We rank occupations along several dimensions: median income in 1950 based on the IPUMS occscore, prestige (Siegel, 1971), cognitive skill requirements (Autor et al., 2003), and leadership and social skill requirements based on O*NET data (Deming, 2017; National Center for O*NET Development, 2026).[6] To capture occupational standing more broadly than earnings alone, we estimate a latent socioeconomic status factor using confirmatory factor analysis and full-information maximum likelihood (FIML) (Enders and Bandalos, 2001). We refer to the resulting standardized measure as the *occ z-score*.[7] Table B.2 reports match-quality comparisons by race.

[5] We conducted this linkage manually by searching census questionnaires on Ancestry before the release of the complete 1950 Census by IPUMS. We subsequently attempted automated linkage using the IPUMS complete-count data, but the match rate was approximately 10 percentage points lower and the accuracy was also lower.

[6] We exclude 390 pilots missing at least one occupational ranking because of missing occupation-level data on occscore, prestige, or skill requirements. The original IPUMS occscore is reported in hundreds of dollars; we convert it to dollars for ease of interpretation. The prestige score ranges from 9.3 to 81.5 in the IPUMS 1950 Census. We standardize the cognitive, leadership, and social skill measures.

[7] We exclude income from the FIML estimation in the pilot sample because the 1950 Census reports income only for a 20% sample, generating too much missingness for stable estimation in this smaller sample.

Second, we collect occupational information from U.S. City Directories for 2,562 pilots over 1960–90. City directories are local publications listing residents' names and occupations. We code the recorded occupations according to the 1950 Census occupational classification system using IPUMS occ1950 codes. We then assign the same occupational rankings as in the 1950 Census—occscore, prestige, and cognitive, leadership, and social skill requirements—and construct an *occ z-score* based on these measures. These data allow us to trace occupational trajectories beyond the immediate postwar period. Table B.3 reports match-quality comparisons by race.

Third, we obtain residential addresses for 2,751 pilots from the U.S. Index to Public Records in the 1990s. This database compiles address information from a variety of public records and allows us to locate pilots late in life at a national scale. We use these addresses to obtain Zillow valuation estimates for each property as of January 2022. For comparison, we use Zillow valuations collected from addresses in sampled ZIP codes. The collection design sampled 10% of ZIP codes and 100 addresses per selected ZIP. We rank pilot valuations against the available comparison observations. This measure has two limitations. First, we do not observe home ownership directly. Second, because household level nationwide valuation data are not available for the 1990s, the ranking is based on 2022 estimated values rather than contemporaneous prices. Our approach therefore assumes that the relative ranking of neighborhoods by housing value is sufficiently persistent between the 1990s and 2022. Table B.4 reports match-quality comparisons by race.

Finally, we link pilots to the public Numerical Identification Files (NUMIDENT) from NARA. The NUMIDENT contains death records for individuals with a verified death or who would have been older than 110 by December 31, 2007. We match these records using name, age, and state of birth, which allows us to recover death ZIP codes for 2,562 pilots. We then merge those ZIP codes to the 2022 IRS Individual Income Tax ZIP Code Data, which report total income and the number of returns by ZIP code. These data allow us to place each pilot's death ZIP code in the national distribution of income per return, providing a measure of neighborhood quality at the end of life.[8] Table B.5 reports match-quality comparisons by race.

[8] Figure B.5 shows the persistence of ZIP code income rankings at death year and in 2022.

Postwar linkage patterns differ across sources and racial groups. White fighter pilots are less likely to be matched than other white pilots, whereas differences by pilot type are small among Black pilots. Black pilots with Southern hometowns are less likely to be matched to Census and housing outcomes, but slightly more likely to be matched to directory and death-income outcomes. Because both pilot type and place of origin may be correlated with later-life outcomes, the regression specifications in Section 4 include fixed effects for pilot type and state of birth.

## 3.2 County-Level Racial Discrimination

Our goal is to measure the discrimination faced by pilots in the counties where they lived after the war. A natural approach would be to use contemporaneous county-level measures of racial discrimination. But this creates an identification problem. Postwar county-level discrimination was not fixed: it evolved alongside migration, local economic change, and the early enforcement of civil rights reforms, all of which could also affect Black socioeconomic outcomes (Derenoncourt, 2022; Bailey et al., 2024). A contemporaneous measure would therefore reflect postwar changes in local discrimination that were themselves related to the outcomes we study.

We therefore use a predetermined proxy for local discrimination: a composite index of pre-1945 racial discrimination at the county level. We construct this index using several related but distinct dimensions of racial discrimination: political support for segregation, racial violence, organized racist activity, residential segregation, and racial inequality in schooling. Following Qian and Tabellini (2024), we summarize six prewar county-level indicators using principal component analysis. These indicators capture political support for the Democratic Party (vote share in presidential and congressional elections, 1900–1930), racial violence (lynchings through 1941), organized racist activity (number of local KKK chapters), residential segregation (Black isolation in 1930 and 1940), and racial inequality in educational attainment (the white-Black education gap in 1940).[9] We standardize each variable, impute missing values iteratively, and use the stan-

[9]Sources: Clubb et al. (2006) for vote data; Bailey et al. (2008) and Seguin and Rigby (2019) for lynching data; Kneebone et al. (2015) for KKK data; and the residential segregation and white-Black education gap measures are computed from the IPUMS full-count censuses.

dardized first principal component as our county discrimination index. This index is intended to capture the persistent component of discrimination rather than short-run postwar variation. It is not a direct measure of the barriers pilots faced in a given year. Instead, it proxies for an underlying local environment of racial hostility and exclusion that likely continued to shape economic opportunity after the war. This interpretation is plausible because racial hostility and segregation were highly persistent across counties, so places with higher pre-war discrimination generally retained it in the postwar decades (Acharya et al., 2016). Our estimates should therefore be interpreted as the effect of settling in counties with higher levels of *historical* discrimination.

Figure 2 maps the discrimination index across counties. High values are concentrated in the South, particularly in Mississippi, Alabama, Georgia, South Carolina, and Louisiana. In contrast, most counties in the Northeast, Midwest, and West show markedly lower index values.

## 3.3 Summary Statistics

Table 1 summarizes the baseline pilot sample and the main variables used in the analysis, all of which are constructed from public data sources.

Black and white pilots in the sample come from similar birth cohorts and have broadly similar schooling levels. White fighter pilots accumulated more aerial victories on average—Tuskegee Airmen were disproportionately assigned to bomber escort missions, which offered fewer opportunities for air-to-air combat (Haulman, 2015).[10] Black pilots are observed across counties with substantial variation in discrimination exposure. For example, based on the county of residence in 1950, the gap between the 10th and 90th percentiles of the discrimination index within the Black pilot sample is 2.7 standard deviations. This dispersion in local environments is central to the analysis below.

## 3.4 Why Pilots?

Before presenting the pilot analysis, we document a broader pattern that helps explain why the pilot setting is especially informative for studying whether dis-

[10] Aerial victory data come from USAAF Historical Research Agency (1978).

crimination is costly. In a large sample of WWII enlistees, Black-white gaps in later-life socioeconomic status widen sharply with measured cognitive ability. This pattern suggests that racial barriers may be especially costly among highly skilled individuals.

We study roughly 442,000 male WWII Army enlistees with Army General Classification Test (AGCT) scores—a wartime cognitive test measure—from the NARA Army Enlistment Records, linked to the 1950 Census and death records; Section A.2 describes the construction of this sample.

Figure 1 presents the primary result. For two outcomes used later in the pilot analysis—1950 socioeconomic status and neighborhood income at death—the Black-white gap increases steeply across the AGCT distribution. In the 1950 SES z-score, for example, the gap rises from roughly 0.2 SDs at low AGCT levels to more than 0.8 SDs near the top of the distribution. The death-ZIP income percentile shows the same qualitative pattern.

This pattern is consistent with the notion that racial barriers imposed especially large losses on high-ability Black men. But the enlistee evidence remains limited because the AGCT captures only one dimension of ability. Even among men with similarly high AGCT scores, substantial differences may remain in other skills relevant for later-life success. AGCT scores are also themselves a reflection of the discrimination to which Black men were exposed when growing up (Bohren et al., 2025).

The Tuskegee Airmen help overcome these limitations. The AAF pilot selection process screened candidates on multiple dimensions and then subjected them to a demanding, standardized training program. Among the men who ultimately earned their wings, ability was therefore likely compressed along more relevant dimensions than in the enlistee sample. The pilot setting thus helps us assess the effects of discrimination with less concern that differences in underlying ability account for the observed outcomes.

# 4. Talent Misallocation Among the Tuskegee Airmen

Did discrimination undermine the economic success of the Tuskegee Airmen? We first document a clear cross-sectional pattern: Black pilots in higher-discrimination counties had worse postwar outcomes, while the corresponding gradient for white

pilots was much weaker. The pattern is shown in Figure 3. In 1950, outcomes for Black pilots decline sharply with county discrimination, both in income and in the ranking of occupations they held, whereas the corresponding gradients for white pilots are markedly flatter.

To estimate this pattern formally, we use the following specification:

$$y_{ic} = \beta_1 \cdot \mathrm{D}_{c(i)} + \beta_2 \cdot \mathrm{Black}_i + \alpha \cdot \mathrm{Black}_i \times \mathrm{D}_{c(i)} + \theta' \mathbf{x}_i + \varepsilon_{ic} \tag{1}$$

where $y_{ic}$ is a postwar socioeconomic outcome for pilot $i$ residing in county $c$ when the outcome is measured, and $D_{c(i)}$ is the discrimination level of that county. A natural starting point would be to regress Black pilots' outcomes directly on county discrimination. But such a regression would be difficult to interpret, because the discrimination index may be correlated with other county characteristics that affect postwar outcomes for both races. We therefore use white pilots as a benchmark group. The coefficient on county discrimination captures the relationship between county characteristics and white pilots' outcomes, while the interaction between the Black indicator and county discrimination captures whether this gradient is systematically different for Black pilots. Our coefficient of interest is therefore $\alpha$. The baseline specifications include fixed effects for pilot type, state of birth, and birth year. We also include state-of-residence fixed effects to compare pilots across counties within the same state. Standard errors are clustered at the state-of-residence level.

## 4.1 Baseline Results

Columns 1–6 in Table 2 show that higher county discrimination is associated with lower occupational rankings for Black pilots relative to white pilots. In Column 1, a one SD increase in county discrimination is associated with $1,009 lower occupational median income for Black pilots relative to white pilots, comparable to the median income gap between a registered nurse and a waitress. The pattern is similarly strong across occupational measures: in more discriminatory counties, Black pilots worked in occupations with lower prestige and lower cognitive, leadership, and social skill requirements. Column 6 summarizes these dimensions in a composite occupational z-score and shows a decline of 0.9 SDs per one

SD increase in discrimination.

The 1950 results are visible in subsequent decades. Column 7 in Table 2 shows that the same pattern repeats in the occupational z-score from city directories in 1960–90. Table 3 shows that these differences persist into late life when outcomes are measured using house values and the income ranking of the ZIP code at death.

One concern about the baseline estimates is selective sorting on residual ability within the Tuskegee pilot sample. If higher-ability Black pilots disproportionately settled in lower-discrimination counties, the cross-sectional estimates could overstate the effect of discrimination. We assess this concern in two steps.

For such sorting to matter quantitatively, residual variation in ability within the pilot sample would need to be substantial. Figure 4 compares the coefficient of variation in years of education in the 1940 Census, an imperfect yet available proxy for ability, between pilots and other groups. Among Black pilots, the coefficient of variation is 0.21, far below that of the general Black population (0.78), lower than that of Black lawyers and judges (0.30), and comparable to that of Black physicians (0.22) and professors (0.20). The same pattern holds for white pilots, whose education dispersion is also far below that of the general white population and close to that of highly selected professions. These comparisons are consistent with the highly selective pilot training process having sharply compressed differences in underlying ability.

We next examine whether higher-ability Black pilots were more likely to settle in lower-discrimination counties. Because ability is not directly observed, we use three proxies: schooling in 1940, schooling at enlistment, and raw aerial victories.[11]

If selective sorting were important, Black pilots with higher levels of these proxies should, relative to white pilots, be disproportionately located in less discriminatory counties. Figure 5 plots the three proxies against the discrimination level of pilots' residence counties in 1950, overlaying Black and white pilots in each plot. The figure does not show this pattern: the relationship for Black pilots is generally flat or slightly positive. Figure B.6, Figure B.7, and Figure B.8 show

[11] Aerial credit data of fighter pilots come from USAAF Historical Research Agency (1978). The raw credit count reflects both wartime performance and unequal flight opportunity for Black pilots (Haulman, 2015).

similar trends if we use the residence counties in later periods. These patterns do not align with the possibility that higher-ability Black pilots systematically sorted into lower-discrimination counties.

## 4.2 Additional Robustness

*Controlling for Prewar County Characteristics and Their Interactions with Race.* The pre-1945 discrimination index proxies for persistent racial hostility, but it is also correlated with regional development. The white-pilot benchmark absorbs county characteristics that affected both races similarly; the remaining concern is that local development may have affected Black and white pilots differently. We therefore add the 1940 county occupational income score, urban share, manufacturing employment share, and high-school graduation share, together with interactions between each characteristic and the Black indicator (Table B.6 and Table B.7). The Black-discrimination coefficients keep the same sign and remain statistically significant for the main summary outcomes, although some estimates attenuate. This pattern suggests that the baseline results are not driven solely by race-specific returns to general county development.

*Race-Specific Fixed Effects.* The baseline specification imposes the same birth-state, birth-year, and pilot-type fixed effects on Black and white pilots. We relax this restriction by interacting each of these fixed effects with the Black indicator. The Black–discrimination coefficients remain negative and statistically significant (Table B.8 and Table B.9).

*CEM for Black–White Comparability.* The baseline specification controls for observable characteristics, but Black and white pilots may still differ in the support of covariates that are difficult to summarize parametrically. We therefore use coarsened exact matching as a diagnostic robustness check (Iacus et al., 2012). In this exercise, the matching indicator is the Black indicator: we match pilots within cells defined by residence county at the time the outcome is measured, education at enlistment, aerial victories, and pilot type, and then re-estimate Equa-

tion 1 using the CEM weights.[12] Table B.10 and Table B.11 show that the estimates remain qualitatively similar in the matched sample.

*Within-State CEM for High-Discrimination Exposure.* A separate concern is that the continuous discrimination gradient may partly compare pilots living in different state environments. We address this concern by defining high-discrimination exposure within each state. For each outcome source, a pilot is assigned to the high-discrimination group if the relevant county discrimination index is above the state mean. The CEM step then matches high- and low-discrimination observations exactly on state, race, and pilot type, and coarsely on median bins of the 1940 county occupational income score, urban share, manufacturing employment share, and high-school graduation share. We estimate a pooled regression of each outcome on Black, high-discrimination exposure, and their interaction using the CEM weights; the regression also includes the original prewar county characteristics interacted with Black. Table B.12 shows that the interaction estimates are negative across the main outcomes.

*Generalized Random Forest.* We also use a generalized random forest following Athey et al. (2019) to assess which county characteristics best explain heterogeneity in the Black-white occ z-score in 1950. The discrimination index ranks among the most important predictors, alongside the local Black population share and ahead of standard economic fundamentals (see Figure B.9). These results support the interpretation that the baseline estimates reflect the racial environment rather than general county economic conditions.

# 5. Evidence Beyond Cross-Sectional Comparisons

The decision to move postwar may reflect ability itself. The low variability and lack of correlation of moving decisions with education make this less likely. Nonetheless, we add two complementary steps. First, we construct an instrument for the discrimination level of 1950 residence counties using the quasi-random ge-

[12] Education is defined as an indicator for having completed at least one year of college. Aerial victories are coded as an indicator for being above or below the median. The control vector includes state of residence fixed effects.

ographic origins of wartime squadron peers. Second, we use pilots who later moved across counties to examine whether changes in the discrimination exposure were associated with changes in occupational rankings.

## 5.1 Instrumental Variable: Peer Origins and Initial Discrimination Exposure

We use the discrimination level of origin counties of a pilot's wartime squadron peers to instrument the racial environment that pilot faced when first settling after the war. The identifying idea is simple: conditional on race and pilot type, peer birth counties were as good as random, but peer origins still shaped initial postwar settlement through wartime social ties.

### 5.1.1 Instrument Construction

We construct the instrument from the birth county discrimination levels of a pilot's wartime squadron peers. Let $\mathcal{P}_i$ denote the set of pilot $i$'s peers in the same flight squadron, excluding pilot $i$.[13] Let $D_{b_j}$ be the pre-1945 discrimination index in peer $j$'s birth county, and let $d_{ij}$ be the distance between pilot $i$'s birth county and peer $j$'s birth county, measured in 100-kilometer units. The peer-origin instrument is

$$Z_i = \frac{\sum_{j \in \mathcal{P}_i} w_{ij} D_{b_j}}{\sum_{j \in \mathcal{P}_i} w_{ij}}, \quad w_{ij} = K(d_{ij}), \quad K(d) = \frac{1}{1+d}. \tag{2}$$

The weights give more weight to peers born nearer to pilot $i$'s birth county, using the same inverse-distance kernel in the zero-stage analysis below.[14] We standardize $Z_i$ before estimation.

To make the idea concrete, consider Elmer Gordon and Raymond Williams. Both were Black fighter pilots born near Norfolk, Virginia, but they served in different squadrons and were therefore exposed to different wartime peer geographies. In Gordon's 100th Fighter Squadron, 67% of leave-one-out peers were born

[13] A squadron includes 100 pilots on average (see Figure B.10 for details).

[14] For Black pilots born outside former Confederate states, we exclude peers born in former Confederate states when constructing $Z_i$. This restriction prevents their instrument from being driven by peer origins in a region to which they rarely moved after the war: among pilots born outside former Confederate states, only 4.7% of Black pilots and 7.1% of white pilots lived in former Confederate states in 1950.

in counties in the top quartile of the discrimination distribution; one peer was born in Richmond City, Virginia, where Gordon lived in 1950, a county near the top of the distribution. In Williams's 302nd Fighter Squadron, the corresponding top-quartile share was lower, at 57%; one peer was born in Hudson County, New Jersey, a county outside the top quartile (67th percentile). This is where Williams chose to settle by 1950. Together, the cases show the basic intuition of the instrument: wartime peer networks linked them to different parts of the country's racial geography after the war.

This instrument relies on two conditions. First, geographic origins of peers in the same flight squadron are as good as random. This assumption is plausible because wartime pilot assignment was constrained only at a broad level by race and pilot type, while high attrition and the pace of mobilization left substantial scope for chance in who ultimately graduated together and served in the same squadron. As a diagnostic for this assignment assumption, we examine whether pilots assigned to the same squadron are unusually similar in their predetermined birth-county characteristics. For each characteristic, we standardize the pilot's own birth-county value and the leave-one-out mean of the same characteristic among his squadron peers. We then regress the standardized own value on the standardized peer mean, absorbing race $\times$ pilot type fixed effects. The sorting statistic is the absolute robust $t$-statistic on the peer-mean coefficient. To form the placebo distribution, we randomly permute squadron labels within race $\times$ pilot type cells, recompute the leave-one-out peer means, and re-estimate the same regression. Table B.13 reports the observed coefficient, the observed sorting statistic, the mean placebo statistic, and the permutation $p$-value, defined as the share of placebo statistics at least as large as the observed one. Across the reported 1940 birth-county characteristics, all permutation $p$-values exceed 0.10.

As a balance check, Table B.14 reports regressions of pilot-level ability proxies on the peer-origin instrument and its interaction with Black status. The specifications include the same fixed effects used in the IV regressions below: residence state, birth state, pilot type, and birth year. The estimates show little relationship with schooling or with aerial victories.

Second, peer origins affected where pilots initially settled after the war. We refer to this as the zero-stage analysis. Wartime social ties could transmit information about hometowns, job opportunities, and local conditions, making some

destinations more salient at the time of discharge. We examine this channel using the following pilot-county specification:

$$\mathbf{1}[\text{pilot } i \text{ lives in } c \text{ in 1950}] = \beta_1 s_{i,c} + \beta_2 K(d_{b_i,c}) + \alpha_i + \alpha_c + \varepsilon_{i,c}, \tag{3}$$

where the dependent variable equals one if pilot $i$ resides in county $c$ in 1950. The variable $s_{i,c}$ is the leave-one-out share of $i$'s squadron peers born in county $c$, and $K(d_{b_i,c})$ is the same inverse-distance kernel defined in Equation 2, now applied to the distance between pilot $i$'s birth county $b_i$ and candidate residence county $c$. Pilot fixed effects absorb individual-specific propensity to move, while county fixed effects absorb common destination attractiveness.

Table B.15 shows that the leave-one-out share of squadron peers born in a county affects whether a pilot settled there in 1950. If the share of a pilot's peers born in a county increases by one percentage point, the pilot's probability of living in that county in 1950 increases by 0.03 percentage points, roughly equal to the baseline mean probability of living in any given county in the pilot-county panel (Column 3).

### 5.1.2 2SLS Estimates

We use the instrument above to estimate the effect of county discrimination on postwar outcomes in 1950. We instrument for $D_i$ and $Black_i \times D_i$ using $Z_i$ and $Black_i \times Z_i$, absorbing fixed effects for pilot type, residence state, birth year, and birth state. The second stage is:

$$y_i = \beta_1 (\widehat{Black_i \times D_i}) + \beta_2 \hat{D}_i + \beta_3 Black_i + \theta' \mathbf{x}_i + \varepsilon_i. \tag{4}$$

The coefficient of interest is $\beta_1$, which captures the effect of local discrimination on Black pilots compared to white pilots.

Table 4 reports the first stage. With two endogenous regressors and two instruments, the relevant diagnostic is the Sanderson-Windmeijer (SW) multivariate $F$, which tests whether the instruments identify each endogenous regressor conditional on the other first stage. This matters because a single strong instrument can make both naive first-stage $F$-statistics look adequate even when the two endogenous variables cannot be separately identified. In the baseline

columns that cluster standard errors at the residence state level, the SW $F$-statistics are 15.48 for discrimination and 20.13 for its interaction with race. When we instead cluster at the flight squadron level, allowing residual shocks to be correlated within wartime peer groups, the corresponding SW $F$-statistics are 15.87 and 27.45.

Table 5 reports the 2SLS estimates. Relative to white pilots, Black pilots in higher discrimination counties in 1950 had worse occupational outcomes. For example, a one SD increase in discrimination lowers the 1950 occupational z-score of Black pilots by 0.86 SDs relative to white pilots. Panel B reports the reduced-form estimates, in which outcomes are regressed directly on the interaction between the Black indicator and the instrument.

Figure B.11 compares the 2SLS estimates between the full sample and compliers. Compliers are defined as pilots whose realized 1950 residence county is among the five candidate counties with the highest predicted settlement probabilities from the peer origin share and the inverse distance kernel in Equation 3. Across outcomes, the estimates for compliers are around 69–110% of those for the full sample.

We compare 2SLS with OLS estimates in Figure B.12. The ratios between 2SLS and OLS estimates cluster around one across different outcomes in 1950.[15] This aligns with the sorting-proxy tests above in Figure 5: there is little indication that OLS is inflated by higher-ability Black pilots sorting into lower-discrimination counties.

### 5.1.3 Robustness

*Peer-Origin Employment Opportunities.* Peers who help pilots settle in less discriminatory counties may also connect them to better jobs. The instrument would then capture the effect of better employment opportunities along with that of discrimination, and the IV estimates would attribute both to discrimination. To address this concern, we control for the weighted mean 1940 occupational score in peers' birth counties and its interaction with Black. The score uses Black residents' occupational scores for Black pilots and white residents' scores for white pilots. Table B.16 reports the estimates. The Black–discrimination coefficients

[15] Figure B.13 repeats this comparison for compliers.

remain negative across outcomes, and for the occupational z-score the coefficient becomes slightly more negative. The negative discrimination differential remains after controlling for this proxy for peer-origin employment opportunities.

*Race-Specific Fixed Effects.* For Black pilots, the instrument varies mostly with their own birthplace, because both the distance weights and the exclusion of peers born in former Confederate states depend on it. We therefore interact the birth-state, birth-year, and pilot-type fixed effects with the Black indicator in both stages. Table B.17 reports the estimates. The coefficient for the occupational z-score is $-0.90$, close to the baseline estimate of $-0.86$.

*First-Stage Heterogeneity.* We test whether the peer-origin instrument affects discrimination exposure more strongly among higher-ability pilots. For Black and white pilots separately, we interact the instrument with each continuous ability proxy. Figure B.14 shows the interaction coefficients. We find no consistent evidence that the first-stage relationship strengthens with measured ability.

*Inverse-Distance Weighting.* One concern with the inverse-distance weighting is that peers born in the same county as pilot $i$ receive relatively high weight in the peer-origin instrument. This concern is limited in scope: 22.4% of the IV analysis sample have any same-county squadron peer. Among all pilots, the mean same-county peer weight share is only 3.5%, with a median of 0.0%. No pilot's instrument is based only on same-county peers.

These results strengthen a causal interpretation of the 1950 cross-sectional relationship. The remaining limitation is that the instrument is informative only for initial postwar exposure. By the 1960s and later, many pilots had moved. The postwar mobility that limits the IV strategy at longer horizons motivates our second approach: the mover design.

## 5.2 Mover Design: Changes in Discrimination Exposure Over Time

The mover design asks whether Black pilots who moved to counties with lower discrimination experienced larger improvements in socioeconomic status (Finkel-

stein et al., 2016; Chetty et al., 2013).

We examine pilots observed both in the 1950 Census and in the 1960–1990 City Directories, where occupational standing can be measured consistently in both periods using the occupational z-score. Among the 1,747 pilots linked in both sources, 72% resided in a different county in the later period, indicating substantial mobility after the war.

Consider George Bolling and William Holloman, who lived in Jackson County, Mississippi, and worked as lodging-house keepers in 1950. Later city-directory records place Bolling in Santa Clara County, California, where measured discrimination exposure was 2.21 SDs lower than in Jackson County. His later occupation was funeral director, with median earnings of $3,100. Holloman's later record is in King County, Washington, where the corresponding decline was smaller, at 1.65 SDs; his listed occupation, machinist apprentice, had median earnings of $2,400. This is the margin in the mover design: Bolling's discrimination exposure fell more, and his occupational upgrade was larger as well. Figure 6A generalizes this comparison: among Black movers, larger declines in discrimination exposure are associated with larger gains in occupational standing.

We estimate the following equation for Black pilot movers $i$:

$$\Delta y_i = \beta \Delta D_i + \gamma X_i + \varepsilon_i, \tag{5}$$

where $\Delta y_i$ (i.e. $y_{i,t+1} - y_{i,t}$) is the change in pilot $i$'s outcome, and $\Delta D_i$ is the corresponding change in county discrimination exposure. We cluster standard errors at the origin county level.

Figure 6B shows the estimate of $\beta$. In the baseline specification, a one SD smaller decline in discrimination exposure is associated with about a 0.5 SD smaller increase in occupational z-score for Black pilots. Tuskegee pilots moving from certain origin counties or to certain destination counties may systematically experience different trajectories, so we plot results of including origin or destination county fixed effects in Figure 6B.

### 5.2.1 Robustness

*Placebo Tests.* A first concern is that changes in discrimination exposure may be correlated with changes in unobservables that also drive occupational growth.

As a placebo check, we estimate the same relationship for white pilots. We do not find this relationship between changes in discrimination and changes in occupational standing for white pilots in Figure 6A. Estimating Equation 5 for white pilots also yields the estimates of $\beta$ close to zero (see Figure 6B).

*Selection over $\Delta D_i$.* If higher ability pilots are both more likely to move to lower-discrimination counties (i.e., smaller $\Delta D_i$) and experience more upward trajectories, the estimate of $\beta$ in Figure 6B would be downward-biased. To test this, we separately regress $\Delta D_i$ against each ability proxy among Black movers: schooling in 1940, schooling at enlistment, and aerial victories. Figure B.15 displays the beta coefficients and 95% confidence intervals. The coefficients are small and statistically indistinguishable from zero, suggesting that residual ability variation among Tuskegee Airmen is not systematically related to $\Delta D_i$.

Taken together, the peer-origin IV evidence for initial settlement and the within-pilot mobility patterns point in the same direction: local racial environments constrained the occupational trajectories of highly skilled Black pilots.

# 6. Conclusion

We study the postwar lives of the Tuskegee Airmen, a cohort of Black pilots whose abilities were screened and standardized through one of the most demanding selection processes of WWII. Tracing them through a wide range of postwar records, we show that Black pilots who settled in higher discrimination counties were sorted into worse occupations at the start of civilian life and continued to experience lower socioeconomic outcomes over subsequent decades.

Early exposure to discrimination had adverse repercussions even for the most talented Black men of their generation. It redirected them into worse career trajectories (Bruhn et al., 2025). Despite living through the dismantling of Jim Crow and major civil rights reforms, Southern Tuskegee Airmen never caught up with Black peers in less discriminatory places, or with white peers in the South. This is visible in the occupations they sorted into, the incomes they earned, and the homes they lived in.

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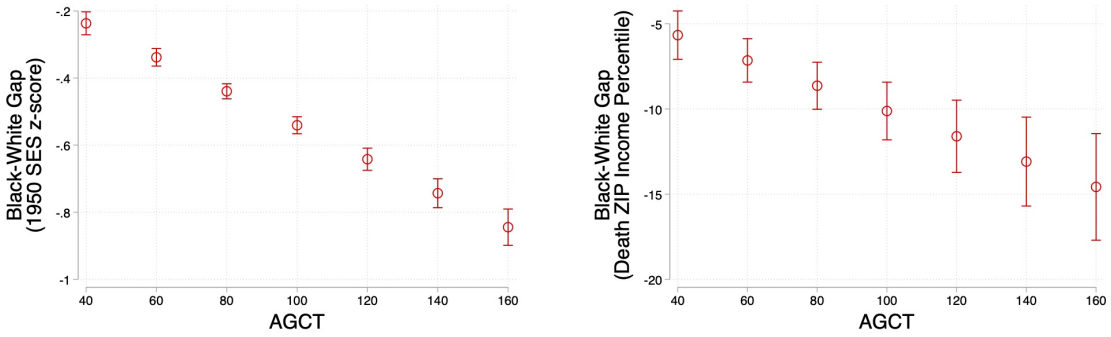
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# Figures

Figure 1: Black-White Gap in Postwar Outcomes by AGCT

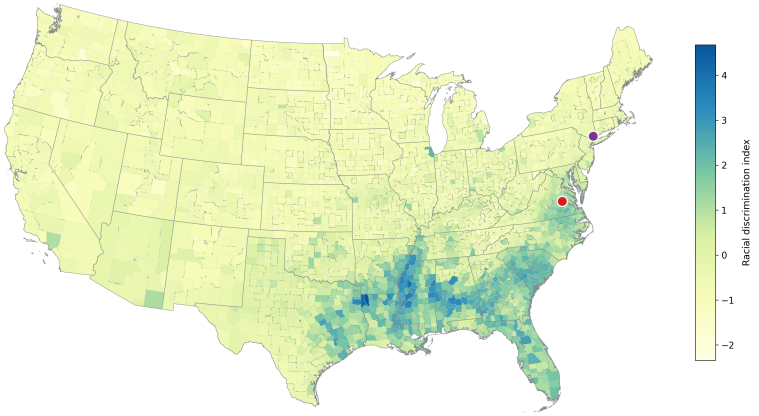



(A) SES z-score (1950)

(B) ZIP code income percentile (death)

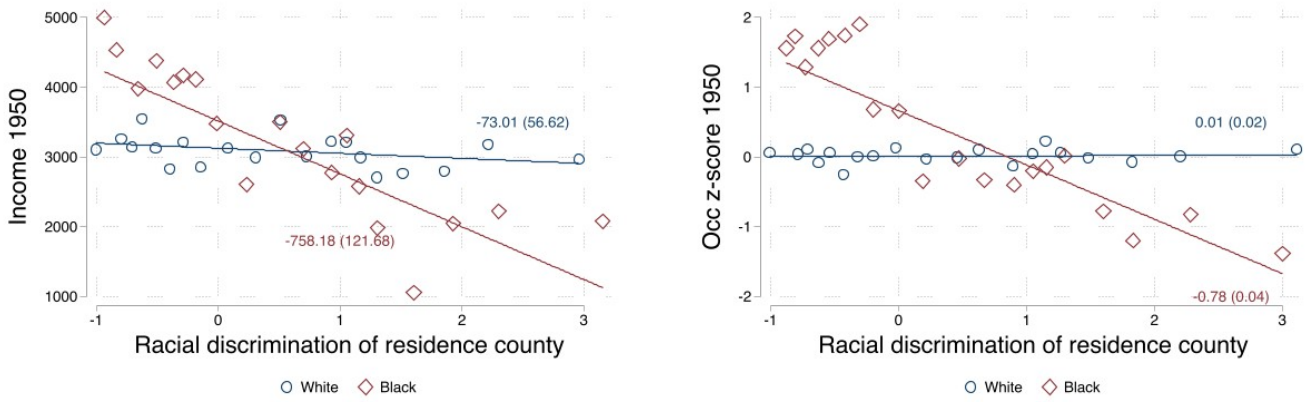
Notes: Each figure shows how the estimated difference in a postwar socioeconomic outcome between Black and white enlistees changes with AGCT scores. Panel A reports the SES z-score from the 1950 Census; Panel B reports the income percentile of the death ZIP code from NUMIDENT. First, we regress each outcome on a Black indicator, AGCT score, and their interaction, controlling for residence county at measurement, residence county at enlistment, birth year, education level, enlistment year-month, and enlistment place. Second, we estimate the marginal effects of the Black indicator when the AGCT score changes from 40 to 160. We plot the estimated marginal effect and its 90% confidence interval (*y*-axis) against the AGCT score (*x*-axis). Standard errors are clustered by the county of residence.

FIGURE 2: Racial Discrimination Index




*Notes*: This map shows a county-level pre-1945 racial discrimination index. See Section 3.2 for details. Darker shades indicate higher levels of the index. The red dot marks Richmond City, VA, and the purple dot marks Westchester County, NY, the two counties in the introductory example.

Figure 3: County Discrimination, Income, and Occupation in 1950




(A) Income (1950)

(B) Occ z-score (1950)

Notes: This figure shows binned scatterplots of county-level racial discrimination against postwar outcomes for Black and white pilots. Panel A shows income in 1950. Panel B shows the occupational z-score in 1950. Each panel also reports the coefficient and standard error from separate regressions of the outcome on the discrimination index for Black and white pilots.

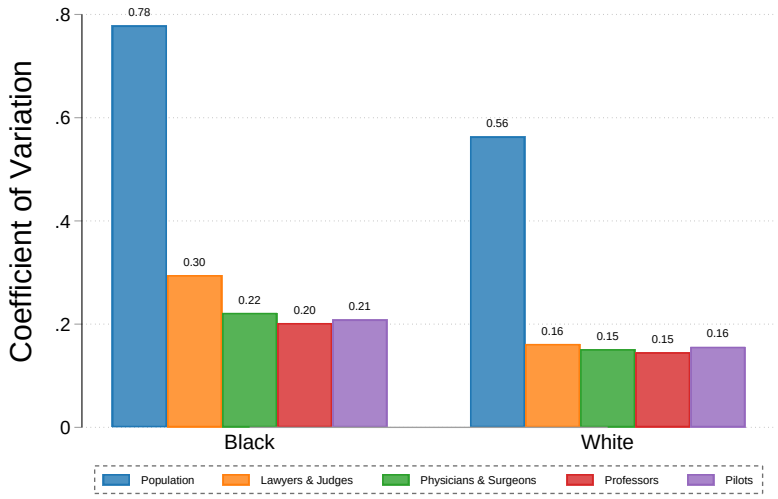
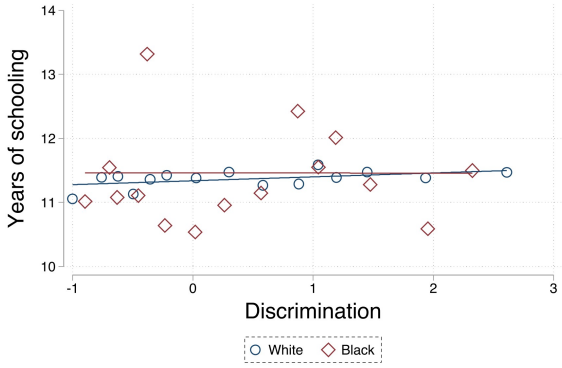


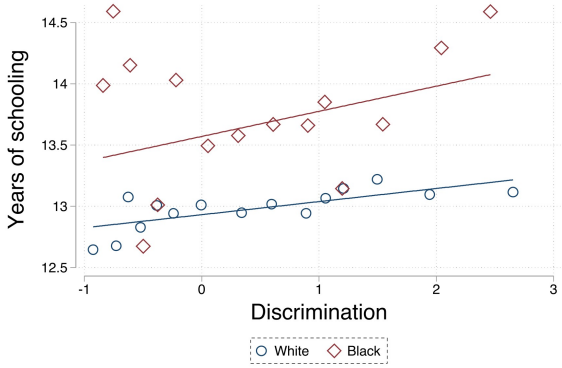

FIGURE 4: Coefficient of Variation in Years of Education by Race and Occupation (1940 Census)

*Notes*: This figure reports the coefficient of variation (CV = SD/mean) of years of education for selected groups, separately for Black and white individuals. "Population" refers to all individuals in the 1940 Census. Lawyers & Judges correspond to occupation code 55 in the 1950 Census occupation classification; Physicians & Surgeons to code 75; Professors to codes 12–29; and Pilots to individuals in our main sample. A lower CV indicates greater homogeneity in educational attainment within a group.

Figure 5: Schooling and Aerial Victories vs. Discrimination of 1950 Residence County




(A) Schooling in 1940



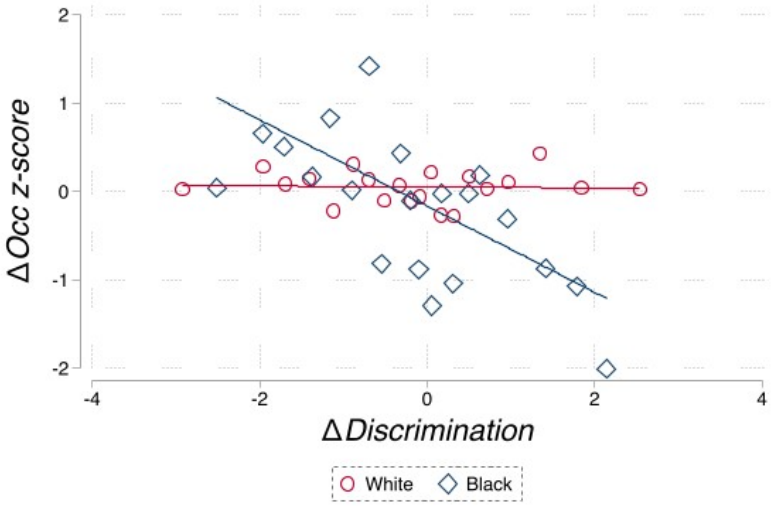

(B) Schooling at enlistment



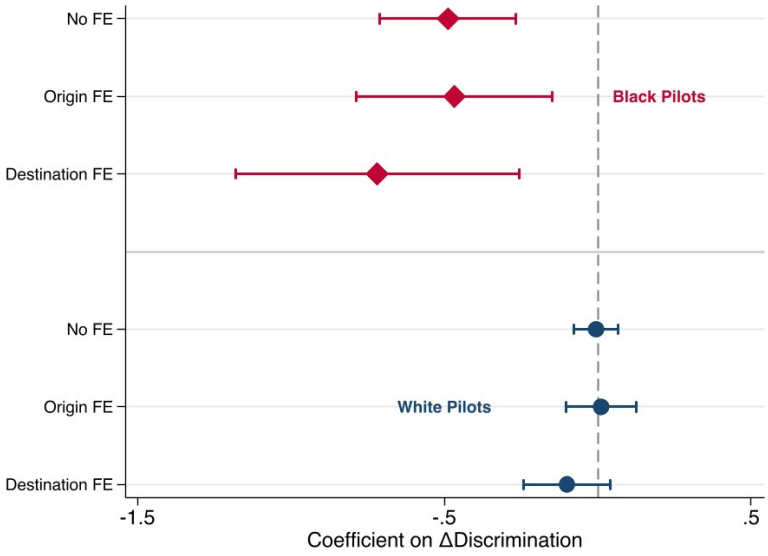

(C) Aerial victories

Notes: Binned scatter plots of years of education in 1940, years of education at enlistment, and raw aerial victories against the racial discrimination index of the 1950 residence county, separately for Black and white pilots. Schooling measures are residualized by year of birth; aerial victories are raw wartime credit counts. Each dot represents the mean within an equal-sized bin; up to 15 bins are used. The lines show the linear fit.

Figure 6: Mover Design Results




(A) Movers from the 1950 Census to 1960–1990 City Directories: ΔD and Δy, by Race




(B) Mover Design Estimates

Notes: Panel 6A shows the change in average $y$ for pilots who moved between counties from the period $t$ (1950 Census) to $t$+1 (1960–1990 City Directories). $y$ is the occ z-score. This figure is the binned scatter plot between ΔD (i.e., $\Delta D = D_{t+1} - D_t$) and Δy (i.e., $y_{t+1} - y_t$) by race. Panel 6B plots corresponding regression estimates and 95% confidence intervals using Equation 5.

# Tables

Table 1: Summary Statistics for Pilots (Public Data)

| | Black | | | | White | | | |
|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Median | N | Mean | SD | Median | N |
| Birth year | 1,921 | 3 | 1,921 | 750 | 1,920 | 2 | 1,921 | 3,577 |
| Aerial victories | 0.21 | 0.63 | 0.00 | 506 | 0.79 | 1.80 | 0.00 | 2,339 |
| Years of schooling (1940 Census) | 11.14 | 2.33 | 11.00 | 291 | 11.39 | 1.78 | 12.00 | 1,507 |
| Years of schooling (Enlistment Records) | 13.72 | 1.63 | 13.00 | 226 | 12.96 | 1.26 | 12.00 | 1,159 |
| Income ($) 1950 | 3,104 | 1,000 | 3,239 | 58 | 3,496 | 1,013 | 3,500 | 361 |
| Occ income score ($) 1950 | 2,881 | 963 | 3,000 | 308 | 3,224 | 997 | 3,200 | 1,865 |
| Occ prestige score 1950 | 43.83 | 15.32 | 40.80 | 308 | 43.95 | 12.92 | 42.50 | 1,865 |
| Occ cognitive score 1950 | -0.00 | 0.91 | -0.34 | 308 | 0.23 | 0.95 | -0.12 | 1,865 |
| Occ leadership score 1950 | -0.21 | 1.09 | -0.10 | 308 | 0.03 | 0.98 | -0.32 | 1,865 |
| Occ social skill score 1950 | -0.63 | 1.08 | -0.82 | 308 | 0.11 | 0.95 | 0.33 | 1,865 |
| SES z-score 1950 | 0.02 | 0.90 | -0.34 | 308 | 0.27 | 0.89 | 0.20 | 1,865 |
| Discrimination 1950 | 0.87 | 0.89 | 0.93 | 308 | 0.35 | 1.06 | 0.07 | 1,865 |
| Occ income score ($) 1960–90 | 2,614 | 1,413 | 2,500 | 354 | 2,992 | 1,201 | 2,600 | 1,999 |
| Occ prestige score 1960–90 | 39.98 | 19.14 | 40.80 | 354 | 44.15 | 16.90 | 40.80 | 1,999 |
| Occ cognitive score 1960–90 | -0.01 | 1.01 | -0.32 | 354 | 0.05 | 0.99 | -0.26 | 1,999 |
| Occ leadership score 1960–90 | 0.15 | 0.89 | 0.11 | 354 | 0.07 | 0.88 | 0.11 | 1,999 |
| Occ social skill score 1960–90 | 0.00 | 0.98 | 0.11 | 354 | 0.06 | 1.00 | 0.10 | 1,999 |
| SES z-score 1960–90 | -0.10 | 1.08 | -0.35 | 354 | 0.09 | 1.00 | -0.26 | 1,999 |
| Discrimination 1960–90 | 0.66 | 0.96 | 0.93 | 354 | 0.29 | 1.06 | -0.02 | 1,999 |
| House percentile 1990s | 51.87 | 32.12 | 57.00 | 438 | 48.35 | 28.03 | 47.00 | 2,313 |
| Discrimination 1990s | 0.67 | 1.00 | 0.85 | 438 | 0.32 | 1.07 | 0.02 | 2,313 |
| ZIP code income percentile death | 52.53 | 35.42 | 58.00 | 383 | 62.99 | 29.32 | 70.00 | 2,179 |
| Discrimination death | 0.67 | 0.95 | 0.93 | 383 | 0.28 | 1.06 | -0.09 | 2,179 |

Notes: See text for variable definitions and data sources.

Table 2: The Effect of Racial Discrimination on Black Pilots' Outcomes (1950 Census and 1960–90 City Directories)

| | 1950 Census | | | | | | 1960-90 City Directories |
|---|---|---|---|---|---|---|---|
| | Income | Prestige | Cognitive | Leadership | Social | Z-score | Z-score |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Black × Discrimination | -1009.269*** | -12.585*** | -0.742*** | -0.244* | -0.733*** | -0.886*** | -0.220*** |
| | (145.398) | (1.544) | (0.099) | (0.130) | (0.074) | (0.098) | (0.076) |
| Black | 480.354*** | 9.639*** | 0.456*** | -0.008 | -0.112 | 0.495*** | -0.105 |
| | (58.246) | (1.145) | (0.092) | (0.163) | (0.135) | (0.063) | (0.101) |
| Discrimination | 166.648*** | 1.736*** | 0.018 | -0.028 | 0.030 | 0.090*** | 0.095** |
| | (39.808) | (0.342) | (0.033) | (0.030) | (0.045) | (0.030) | (0.038) |
| dydx Margin on Black | 50.423 | 4.278*** | 0.140* | -0.112 | -0.424*** | 0.117** | -0.180* |
| | (52.871) | (1.112) | (0.079) | (0.172) | (0.153) | (0.051) | (0.094) |
| Adj. $R^2$ | 0.135 | 0.099 | 0.090 | 0.051 | 0.146 | 0.114 | 0.052 |
| $N$ | 2,148 | 2,148 | 2,148 | 2,148 | 2,148 | 2,148 | 2,529 |

Notes: This table shows regression results using pilot outcomes from the 1950 Census and 1960–90 City Directories. Outcomes are occupation-based: median income in 1950 dollars, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. We regress each outcome on a Black indicator, a continuous measure of the racial discrimination index for the residence county at the time the outcome is measured, and their interaction. All specifications include fixed effects for residence state, birth state, birth year, and pilot type (fighter or bomber). Standard errors (in parentheses) are clustered at the residence state level. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table 3: The Effect of Racial Discrimination on Black Pilots' Outcomes (1990s Index to Public Records and NUMIDENT)

| | House percentile (1) | ZIP code income percentile (2) |
|---|---|---|
| Black × Discrimination | -5.079*** | -8.772*** |
| | (1.756) | (2.357) |
| Black | 1.087 | -7.935*** |
| | (3.109) | (2.690) |
| Discrimination | 5.897*** | 2.168* |
| | (1.334) | (1.218) |
| dydx Margin on Black | -0.827 | -10.938*** |
| | (2.861) | (2.749) |
| Adj. $R^2$ | 0.442 | 0.118 |
| $N$ | 2,715 | 2,529 |

Notes: This table shows regression results using pilot outcomes from the 1990s Index to Public Records and NUMIDENT. We regress each outcome on a Black indicator, a continuous measure of racial discrimination in the residence county when the outcome was measured, and their interaction. All specifications include fixed effects for residence state, birth state, birth year, and pilot type. Standard errors (in parentheses) are clustered at the residence state level. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table 4: First Stage of 2SLS

| | Cluster by residence state | | Cluster by flight squadron | |
|---|---|---|---|---|
| | Discrimination | Black × Discrimination | Discrimination | Black × Discrimination |
| | (1) | (2) | (3) | (4) |
| Z | 0.15*** | -0.06** | 0.15*** | -0.06** |
| | (0.04) | (0.02) | (0.04) | (0.02) |
| Black × Z | -0.10 | 0.33*** | -0.10* | 0.33*** |
| | (0.08) | (0.07) | (0.05) | (0.05) |
| Black | 0.32*** | 0.37** | 0.32*** | 0.37*** |
| | (0.11) | (0.17) | (0.09) | (0.09) |
| SW $F$ | 15.48 | 20.13 | 15.87 | 27.45 |
| $N$ | 2,056 | 2,056 | 2,056 | 2,056 |

Notes: This table shows the first stage of 2SLS, where the endogenous variables are the county racial discrimination index and its interaction with the Black indicator. The instrument is the weighted average discrimination level of squadron peers' birth counties, defined in Equation 2. All specifications absorb fixed effects for residence state, pilot type, birth year, and birth state. Columns (1)–(2) cluster standard errors at the residence-state level; columns (3)–(4) cluster standard errors at the flight-squadron level. Standard errors are reported in parentheses. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table 5: 2SLS and Reduced Form Results

| | Income | Prestige | Cognitive | Leadership | Social | Z-score |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: 2SLS* | | | | | | |
| Black × Discrimination | -1166.12*** | -12.01*** | -0.60*** | -0.16 | -0.82*** | -0.86*** |
| | (160.99) | (3.13) | (0.17) | (0.32) | (0.26) | (0.13) |
| Black | 665.10*** | 8.51*** | 0.28 | -0.12 | -0.05 | 0.45*** |
| | (166.03) | (3.19) | (0.22) | (0.33) | (0.27) | (0.15) |
| Discrimination | 48.43 | 3.34 | 0.19 | 0.01 | 0.04 | 0.16 |
| | (346.55) | (3.07) | (0.31) | (0.28) | (0.28) | (0.24) |
| $N$ | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 |
| *Panel B: Reduced Form* | | | | | | |
| Black × Z | -395.38*** | -4.35*** | -0.22*** | -0.06 | -0.28*** | -0.30*** |
| | (102.65) | (1.62) | (0.08) | (0.11) | (0.09) | (0.09) |
| Black | 253.00 | 5.17* | 0.12 | -0.18 | -0.34* | 0.18 |
| | (164.02) | (2.91) | (0.18) | (0.22) | (0.21) | (0.17) |
| Z | 73.96 | 1.20** | 0.06 | 0.01 | 0.05 | 0.07* |
| | (57.09) | (0.58) | (0.05) | (0.05) | (0.05) | (0.04) |
| $N$ | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 |
| Adj. $R^2$ | 0.046 | 0.021 | 0.034 | 0.049 | 0.099 | 0.030 |

Notes: This table shows 2SLS and reduced form estimates of the effect of racial discrimination on Black pilots' postwar outcomes. Outcomes are occupation-based: median income in 1950 dollars, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. The instrument is the weighted average discrimination level of squadron peers' birth counties, defined in Equation 2. All specifications absorb fixed effects for residence state, pilot type, birth year, and birth state. Standard errors are clustered at the residence-state level and reported in parentheses. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

# Online Appendix

## Grounded: Discrimination and Lifecycle Outcomes of America's First Black High Flyers

October 2026

# A. Appendix Data

## A.1 Data Source on White Pilots

The following list details the sources of our data on white pilots:

1. Fighter Pilots

    - The 4th Fighter Group. Its complete roster of 572 pilots is from the Official Site of the 4th Fighter Group—World War II, which gives each pilot's name, birth date, birth place, death date, death place, etc. We use this website's information to locate pilots in *Ancestry*, and recover birth year and birthplace for 567 pilots.
    - The 55th Fighter Group. The group had 344 pilots who served during WWII according to http://www.55th.org, which provides each pilot's name and hometown. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 337 pilots.
    - The 78th Fighter Group. It has 437 pilots according to (Miller, 2001, pp. 304-309). We use *Duxford Diary* by Raymond Shepard to find each pilot's hometown.[1] We use the hometown information to locate pilots in *Ancestry*, and recover birth year and birthplace for 432 pilots.
    - The 339th Fighter Group. It has 374 pilots according to the *Roster of 339th Fighter Group Personnel* on http://339thfg.com. This website's roster has each pilot's name and hometown. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 372 pilots.
    - The 353rd Fighter Group. The group has 409 pilots according to Graham Cross, who published his data on https://353rdfightergroup.wordpress.com. This list provides each pilot's name and hometown. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 385 pilots.
    - The 358th Fighter Group. It has 294 pilots according to the 358th FG Yearbook, which is available on https://www.fold3.com. This book provides pilots' names and hometowns. We use the information to locate

[1] We thank Curt Shepard for sharing his data with us.

pilots in *Ancestry*, and recover birth year and birthplace for 247 pilots.

- The 359th Fighter Group. It has 365 pilots according to a list that we received from Janet Fogg. It has each pilot's name and hometown. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 314 pilots.
- The 364th Fighter Group. The group has 467 pilots according to (Miller, 2001, pp. 464-469). We use publicly available information to locate pilots in *Ancestry*, and recover birth year and birthplace for 415 pilots.

2. Bombing Pilots

    - The 303rd Bombardment Group. According to the crew list on http://www.303rdbg.com, which provides each pilot's name and hometown, it had 297 (co-)pilots during WWII. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 287 pilots.
    - The 384th Bombardment Group. It has 929 (co-)pilots according to a roster we received from Fred Preller, which includes names and hometowns. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 732 pilots.
    - The 401st Bombardment Group. The group has 654 (co-)pilots according to the combat personnel list on http://401bg.org. In addition, *Directory of Names and Addresses of Former Members of the 401st Bombardment Group* published by the Air Force in 1947 provides pilots' address information. We use the information to locate pilots in *Ancestry*, and recover birth year and birthplace for 553 pilots.

Together, these rosters contain 5,142 white pilots. We recover birth year and birthplace for 4,641 of them: 3,069 fighter pilots and 1,572 bombing pilots. These counts precede the death-year and birth-year restrictions described in the main text.

## A.2 WWII Enlistee Sample and Linkage

Following Ferrie et al. (2012), we use the NARA Army Enlistment Records (NARA-AER) to recover AGCT scores for approximately 442,000 male Army enlistees, of whom 10.5% are Black. Ferrie et al. (2012) show that the "weight" variable in the

NARA-AER stores AGCT scores for men admitted during March–May 1943. We link these enlistees to the 1950 Census using the algorithm developed by Abramitzky et al. (2021), obtaining a linkage rate of 51%. We also link them to NUMIDENT records to measure neighborhood quality at death. Table B.18 compares enlistees with and without AGCT scores. Table B.19 compares Census-matched and unmatched enlistees, and Table B.20 compares NUMIDENT-matched and unmatched enlistees. Table B.21 reports summary statistics for the enlistee sample.

# B. Appendix Figures and Tables

Table B.1: Balance of Matched and Unmatched Black Pilots (Roster to Ancestry)

| | (1) Mean Matched | (2) Mean Unmatched | (3) Difference | (4) $p$-value |
|---|---|---|---|---|
| 2nd Lt at graduation | 0.694 | 0.653 | 0.041 | 0.402 |
| Graduation year | 1943.9 | 1944.1 | -0.175 | 0.103 |
| Fighter pilot | 0.707 | 0.577 | 0.130 | 0.008 |
| Southern state | 0.321 | 0.257 | 0.064 | 0.190 |
| $N$ | 903 | 106 | | |

Notes: This table reports means of key covariates for Black pilots who were matched and unmatched from the Tuskegee Airmen roster to Ancestry records. Column (1) shows the mean for matched pilots, column (2) for unmatched pilots, column (3) the difference, and column (4) the $p$-value from a two-sample $t$-test. "2nd Lt at graduation" is an indicator for whether the pilot held the rank of Second Lieutenant at graduation. "Fighter pilot" indicates assignment as a fighter pilot. "Southern state" indicates that the hometown recorded in military records is in a Southern state.

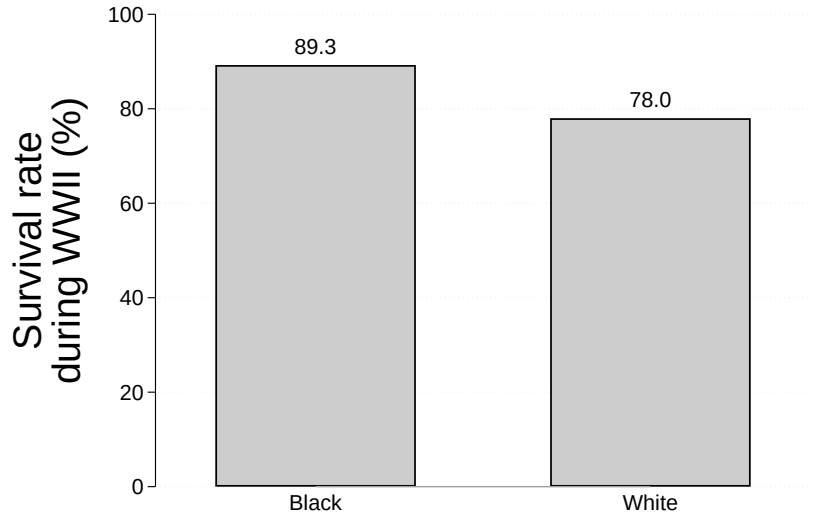



FIGURE B.1: WWII Survival Rates by Race

*Notes*: This figure reports the share of linked Black and white pilots who survived WWII. The sample includes pilots in the raw rosters with birth year and usable birthplace information, matching the first-stage Ancestry linkage definition used in the pilot data construction. Wartime deaths are defined as deaths on or before September 2, 1945; pilots without a recorded death by that date are classified as survivors.

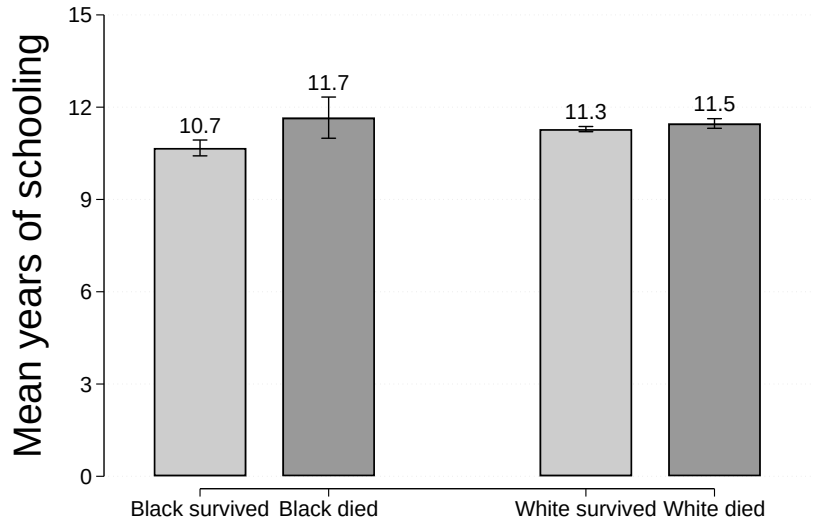



FIGURE B.2: Years of Schooling by Wartime Death Status: 1940 Census

*Notes*: This figure reports mean years of schooling from the 1940 Census by race and wartime death status. Schooling is converted from the IPUMS high-grade variable, residualized on birth year, and shifted back by adding the sample mean. Vertical lines show 95 percent confidence intervals.

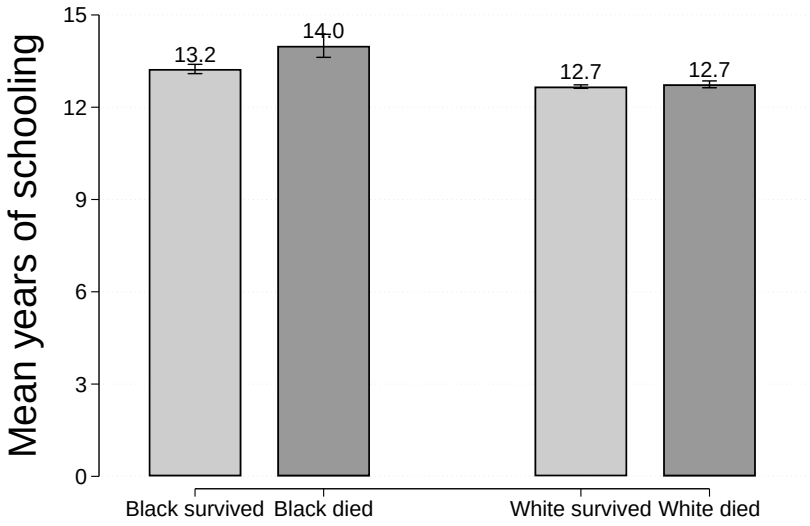



FIGURE B.3: Years of Schooling by Wartime Death Status: Enlistment Records

*Notes*: This figure reports mean years of schooling at enlistment by race and wartime death status. Schooling is residualized on birth year and shifted back by adding the sample mean. Vertical lines show 95 percent confidence intervals.

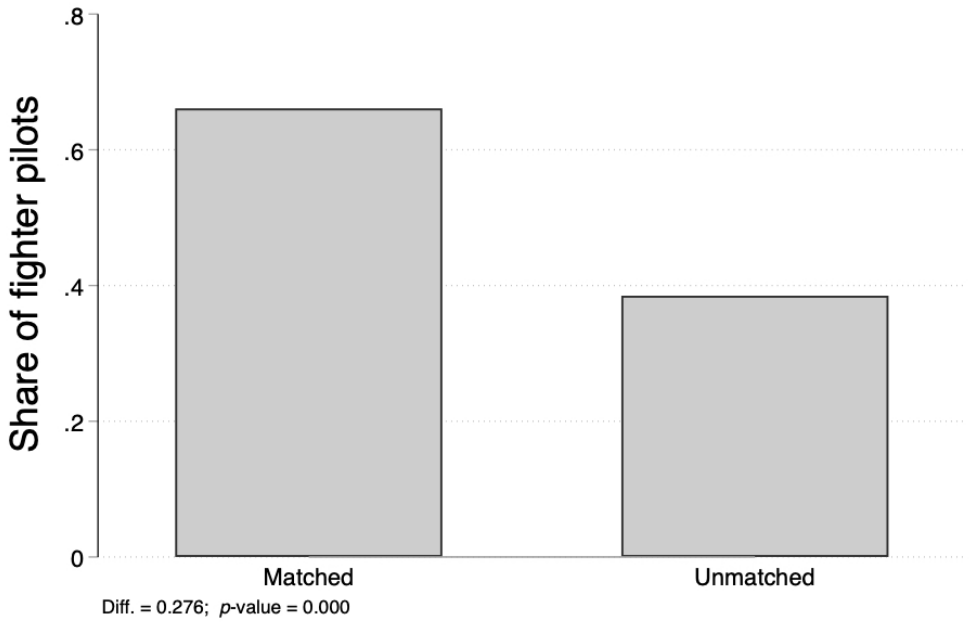



FIGURE B.4: Fighter Pilot Share Among Matched and Unmatched White Pilots (Roster to Ancestry)

*Notes*: This figure shows the share of fighter pilots among white pilots who were matched and unmatched from the roster to Ancestry records. The difference and *p*-value are from a two-sample *t*-test.

Table B.2: Balance of Matched and Unmatched Pilots (Ancestry to 1950 Census)

| | Black | | | | White | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) Mean M | (2) Mean U | (3) Diff | (4) $p$ | (5) Mean M | (6) Mean U | (7) Diff | (8) $p$ |
| 2nd Lt at graduation | 0.670 | 0.685 | -0.016 | 0.648 | | | | |
| Graduation year | 1943.9 | 1943.9 | 0.001 | 0.991 | | | | |
| Fighter pilot | 0.692 | 0.677 | 0.015 | 0.672 | 0.641 | 0.675 | -0.033 | 0.043 |
| Southern state | 0.292 | 0.346 | -0.054 | 0.117 | | | | |
| Birth year | 1920.3 | 1920.8 | -0.411 | 0.046 | 1920.4 | 1920.5 | -0.097 | 0.252 |
| Born in South | 0.497 | 0.538 | -0.041 | 0.267 | 0.277 | 0.277 | 0.000 | 0.995 |
| $N$ | 338 | 412 | | | 2,225 | 1,352 | | |

Notes: This table reports means of key covariates for pilots matched and unmatched from Ancestry records to the 1950 Census, separately for Black and white pilots. We restrict the sample to pilots successfully matched to Ancestry in the first step. Columns (1)–(4) report results for Black pilots; columns (5)–(8) for white pilots. Differences are tested using two-sample $t$-tests. "2nd Lt at graduation" and "Southern state" are reported for Black pilots only. "Born in South" indicates birth in a Southern state.

Table B.3: Balance of Matched and Unmatched Pilots (Ancestry to City Directories)

| | Black | | | | White | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) Mean M | (2) Mean U | (3) Diff | (4) $p$ | (5) Mean M | (6) Mean U | (7) Diff | (8) $p$ |
| 2nd Lt at graduation | 0.677 | 0.679 | -0.002 | 0.947 | | | | |
| Graduation year | 1943.9 | 1944.0 | -0.089 | 0.200 | | | | |
| Fighter pilot | 0.682 | 0.686 | -0.005 | 0.895 | 0.623 | 0.702 | -0.080 | 0.000 |
| Southern state | 0.339 | 0.305 | 0.034 | 0.324 | | | | |
| Birth year | 1920.5 | 1920.6 | -0.081 | 0.694 | 1920.3 | 1920.6 | -0.342 | 0.000 |
| Born in South | 0.533 | 0.505 | 0.028 | 0.453 | 0.279 | 0.274 | 0.005 | 0.752 |
| $N$ | 383 | 367 | | | 2,179 | 1,398 | | |

Notes: This table reports means of key covariates for pilots matched and unmatched from Ancestry records to City Directories, separately for Black and white pilots. We restrict the sample to pilots successfully matched to Ancestry in the first step. Columns (1)–(4) report results for Black pilots; columns (5)–(8) for white pilots. Differences are tested using two-sample $t$-tests. "2nd Lt at graduation" and "Southern state" are reported for Black pilots only. "Born in South" indicates birth in a Southern state.

Table B.4: Balance of Matched and Unmatched Pilots (Ancestry to Index to Public Records)

| | Black | | | | White | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) Mean M | (2) Mean U | (3) Diff | (4) $p$ | (5) Mean M | (6) Mean U | (7) Diff | (8) $p$ |
| 2nd Lt at graduation | 0.701 | 0.646 | 0.055 | 0.114 | | | | |
| Graduation year | 1943.9 | 1943.9 | 0.008 | 0.911 | | | | |
| Fighter pilot | 0.677 | 0.694 | -0.017 | 0.628 | 0.637 | 0.685 | -0.048 | 0.004 |
| Southern state | 0.292 | 0.365 | -0.073 | 0.036 | | | | |
| Birth year | 1920.6 | 1920.6 | -0.053 | 0.797 | 1920.6 | 1920.2 | 0.383 | 0.000 |
| Born in South | 0.499 | 0.548 | -0.050 | 0.183 | 0.270 | 0.290 | -0.020 | 0.210 |
| $N$ | 438 | 312 | | | 2,313 | 1,264 | | |

Notes: This table reports means of key covariates for pilots matched and unmatched from Ancestry records to the Index to Public Records, separately for Black and white pilots. We restrict the sample to pilots successfully matched to Ancestry in the first step. Columns (1)–(4) report results for Black pilots; columns (5)–(8) for white pilots. Differences are tested using two-sample $t$-tests. "2nd Lt at graduation" and "Southern state" are reported for Black pilots only. "Born in South" indicates birth in a Southern state.

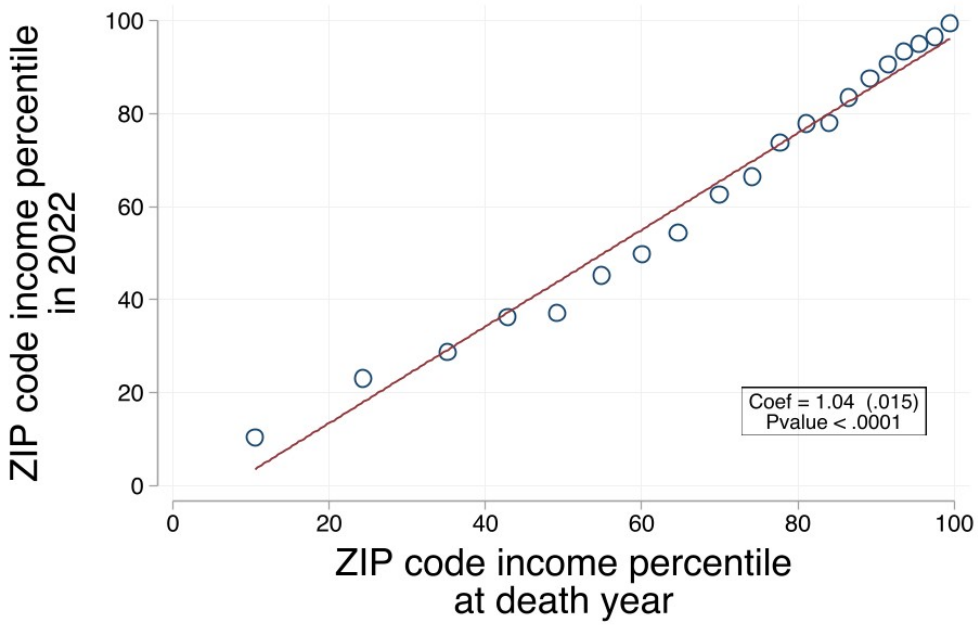



FIGURE B.5: Persistence of ZIP Code Income Rankings: Death-Year vs. 2022 Percentiles

*Notes*: This figure plots the binned relationship between a pilot's death ZIP code income percentile measured in the year of death (horizontal axis) and the same ZIP code's income percentile measured in 2022 (vertical axis). Income percentiles are based on adjusted gross income (AGI) per tax return from IRS Statistics of Income (SOI) ZIP code data, ranked nationally within each year. The fitted line and reported coefficient are from an OLS regression of the 2022 percentile on the death-year percentile. The sample includes pilots from the Social Security Death Index who died in years covered by IRS SOI data (1998–2022) and whose death ZIP code appears in both the death-year and 2022 IRS files.

Table B.5: Balance of Matched and Unmatched Pilots (Ancestry to NUMIDENT)

| | Black | | | | White | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) Mean M | (2) Mean U | (3) Diff | (4) $p$ | (5) Mean M | (6) Mean U | (7) Diff | (8) $p$ |
| 2nd Lt at graduation | 0.677 | 0.679 | -0.002 | 0.947 | | | | |
| Graduation year | 1943.9 | 1944.0 | -0.089 | 0.200 | | | | |
| Fighter pilot | 0.682 | 0.686 | -0.005 | 0.895 | 0.623 | 0.702 | -0.080 | 0.000 |
| Southern state | 0.339 | 0.305 | 0.034 | 0.324 | | | | |
| Birth year | 1920.5 | 1920.6 | -0.081 | 0.694 | 1920.3 | 1920.6 | -0.342 | 0.000 |
| Born in South | 0.533 | 0.505 | 0.028 | 0.453 | 0.279 | 0.274 | 0.005 | 0.752 |
| $N$ | 383 | 367 | | | 2,179 | 1,398 | | |

Notes: This table reports means of key covariates for pilots matched and unmatched from Ancestry records to NUMIDENT, separately for Black and white pilots. We restrict the sample to pilots successfully matched to Ancestry in the first step. Columns (1)–(4) report results for Black pilots; columns (5)–(8) for white pilots. Differences are tested using two-sample $t$-tests. "2nd Lt at graduation" and "Southern state" are reported for Black pilots only. "Born in South" indicates birth in a Southern state.

Table B.6: Robustness Checks Controlling for Prewar County Characteristics (1950 Census and 1960–90 City Directories)

| | 1950 Census | | | | | | 1960-90 City Directories |
|---|---|---|---|---|---|---|---|
| | Income | Prestige | Cognitive | Leadership | Social | Z-score | Z-score |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Black × Discrimination | -1005.138*** | -10.899*** | -0.722*** | -0.297* | -0.736*** | -0.839*** | -0.177** |
| | (137.982) | (1.423) | (0.116) | (0.174) | (0.134) | (0.092) | (0.077) |
| Black | -2099.921** | -32.839* | 1.167 | 0.438 | 0.590 | -0.872 | -1.600* |
| | (962.426) | (16.823) | (1.586) | (1.961) | (1.645) | (1.196) | (0.846) |
| Discrimination | 126.982*** | 1.789*** | 0.050 | 0.081** | 0.051 | 0.103*** | 0.024 |
| | (46.609) | (0.432) | (0.030) | (0.035) | (0.049) | (0.035) | (0.062) |
| 1940 mean occscore | 46.791 | -0.000 | -0.040 | -0.019 | -0.011 | -0.005 | 0.018 |
| | (29.987) | (0.415) | (0.026) | (0.029) | (0.030) | (0.028) | (0.033) |
| Black × 1940 mean occscore | 136.146** | 2.106* | -0.043 | 0.004 | -0.020 | 0.071 | 0.107** |
| | (55.298) | (1.095) | (0.082) | (0.114) | (0.073) | (0.070) | (0.051) |
| 1940 urban share | -21.717 | -0.965 | 0.180 | -0.234 | 0.147 | 0.022 | -0.048 |
| | (206.775) | (2.746) | (0.165) | (0.173) | (0.199) | (0.182) | (0.169) |
| Black × 1940 urban share | -86.529 | -12.480 | -0.339 | -0.098 | -0.227 | -0.479 | -0.303 |
| | (365.786) | (9.522) | (0.604) | (0.682) | (0.650) | (0.481) | (0.323) |
| 1940 manufacturing share | -439.595 | -2.244 | -0.471 | -1.094** | -1.125* | -0.508 | -0.458 |
| | (507.464) | (6.788) | (0.433) | (0.517) | (0.572) | (0.467) | (0.586) |
| Black × 1940 manufacturing share | -1105.083 | -20.435 | 1.277* | 1.020 | 0.895 | -0.208 | -1.385* |
| | (805.267) | (17.456) | (0.757) | (1.639) | (1.538) | (0.790) | (0.807) |
| 1940 high-school graduate share | -507.482 | 5.111 | -0.089 | -0.081 | 0.024 | -0.020 | 2.147*** |
| | (487.236) | (5.858) | (0.587) | (0.790) | (0.629) | (0.488) | (0.780) |
| Black × 1940 high-school graduate share | -1212.636 | 14.893 | 0.789 | -1.989 | -0.777 | 0.113 | -1.783 |
| | (861.490) | (13.850) | (1.109) | (2.819) | (2.708) | (0.957) | (1.360) |
| dydx Margin on Black | -26.775 | 3.981*** | 0.181* | -0.061 | -0.389*** | 0.112* | -0.223*** |
| | (60.678) | (1.152) | (0.102) | (0.131) | (0.125) | (0.061) | (0.084) |
| Adj. $R^2$ | 0.139 | 0.105 | 0.095 | 0.070 | 0.151 | 0.113 | 0.074 |
| $N$ | 2,148 | 2,148 | 2,148 | 2,148 | 2,148 | 2,148 | 2,529 |

Notes: Outcomes are occupation-based: median income in 1950 dollars, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. This table shows the robustness of our results in Table 2 by controlling for the 1940 county occupational income score, urbanization, manufacturing share, and schooling (and their interactions with race). * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table B.7: Robustness Checks Controlling for Prewar County Characteristics (1990s Index to Public Records and NUMIDENT)

| | House percentile | ZIP code income percentile |
|---|---|---|
| | (1) | (2) |
| Black × Discrimination | -4.543** | -6.694*** |
| | (2.212) | (2.270) |
| Black | -41.713** | -22.370 |
| | (17.410) | (29.315) |
| Discrimination | 3.093 | 0.034 |
| | (1.997) | (2.131) |
| 1940 mean occscore | 1.480 | 1.115 |
| | (1.296) | (0.768) |
| Black × 1940 mean occscore | 2.868* | 0.804 |
| | (1.688) | (1.673) |
| 1940 urban share | -1.422 | -5.492 |
| | (5.164) | (6.321) |
| Black × 1940 urban share | 6.252 | -16.195 |
| | (9.304) | (9.944) |
| 1940 manufacturing share | -22.255 | -10.928 |
| | (25.988) | (16.390) |
| Black × 1940 manufacturing share | -74.335** | -9.211 |
| | (34.388) | (25.466) |
| 1940 high-school graduate share | 65.994** | 71.610*** |
| | (25.465) | (19.609) |
| Black × 1940 high-school graduate share | -44.267 | 17.679 |
| | (57.096) | (28.346) |
| dydx Margin on Black | -3.222** | -11.292*** |
| | (1.625) | (2.947) |
| Adj. $R^2$ | 0.492 | 0.150 |
| $N$ | 2,715 | 2,529 |

Notes: This table shows the robustness of our results in Table 3 by controlling for the 1940 county occupational income score, urbanization, manufacturing share, and schooling (and their interactions with race). * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table B.8: Robustness Checks with Race-Specific Fixed Effects (1950 Census and 1960–90 City Directories)

| | 1950 Census | | | | | | 1960-90 City Directories |
|---|---|---|---|---|---|---|---|
| | Income | Prestige | Cognitive | Leadership | Social | Z-score | Z-score |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Black × Discrimination | -983.633*** | -11.482*** | -0.758*** | -0.259* | -0.749*** | -0.859*** | -0.237*** |
| | (122.813) | (1.577) | (0.091) | (0.142) | (0.091) | (0.091) | (0.063) |
| Discrimination | 163.775*** | 1.687*** | 0.023 | -0.026 | 0.029 | 0.090*** | 0.098** |
| | (40.190) | (0.358) | (0.034) | (0.028) | (0.043) | (0.031) | (0.038) |
| Black × Predetermined FE | Yes | Yes | Yes | Yes | Yes | Yes | Yes |
| Adj. $R^2$ | 0.120 | 0.093 | 0.078 | 0.067 | 0.149 | 0.100 | 0.056 |
| $N$ | 2,140 | 2,140 | 2,140 | 2,140 | 2,140 | 2,140 | 2,525 |

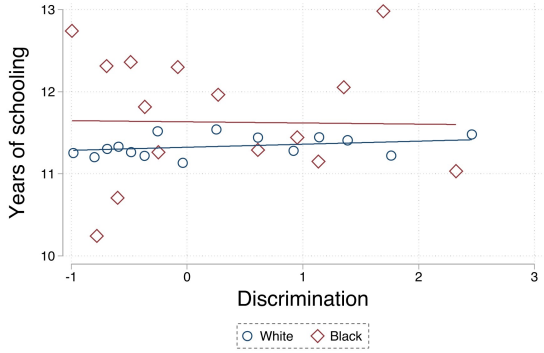
Notes: Relative to Table 2, specifications interact the birth-state, birth-year, and pilot-type fixed effects with Black. All other details follow Table 2.

Table B.9: Robustness Checks with Race-Specific Fixed Effects (1990s Index to Public Records and NUMIDENT)

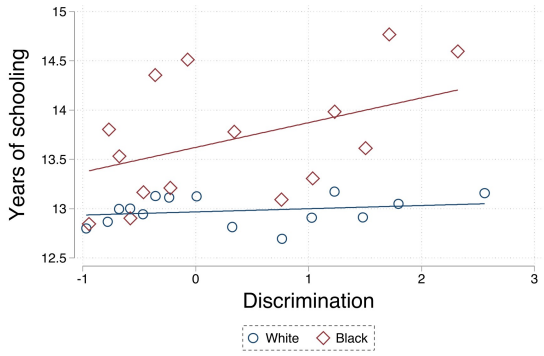
| | House percentile | ZIP code income percentile |
|---|---|---|
| | (1) | (2) |
| Black × Discrimination | -3.938** | -9.302*** |
| | (1.651) | (2.391) |
| Discrimination | 5.890*** | 2.371** |
| | (1.272) | (1.173) |
| Black × Predetermined FE | Yes | Yes |
| Adj. $R^2$ | 0.450 | 0.117 |
| $N$ | 2,708 | 2,525 |

Notes: Relative to Table 3, specifications interact the birth-state, birth-year, and pilot-type fixed effects with Black. All other details follow Table 3.

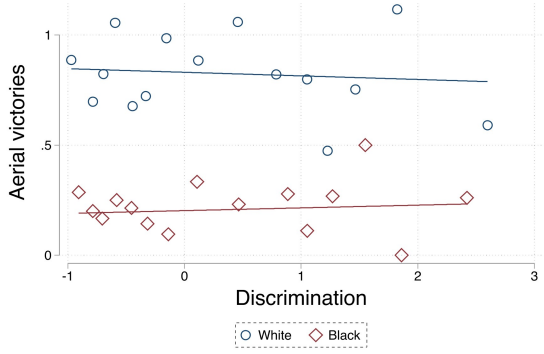
Figure B.6: Schooling and Aerial Victories vs. Discrimination of Residence County in 1960–90 City Directories




(A) Schooling in 1940



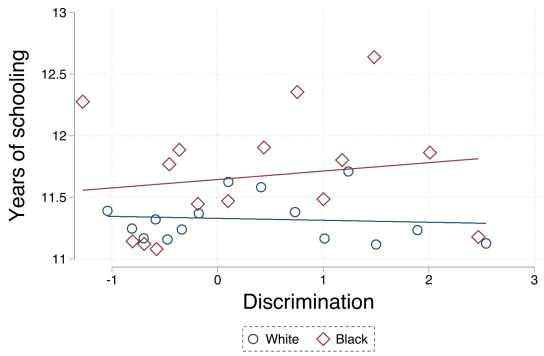

(B) Schooling at enlistment



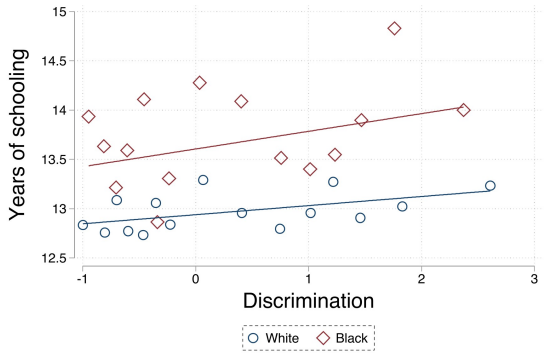

(C) Aerial victories

Notes: Binned scatter plots of years of education in 1940, years of education at enlistment, and raw aerial victories against the racial discrimination index of the residence county in 1960–90 City Directories, separately for Black and white pilots. Schooling measures are residualized by year of birth; aerial victories are raw wartime credit counts. Each dot represents the mean within an equal-sized bin; up to 15 bins are used. The lines show the linear fit.

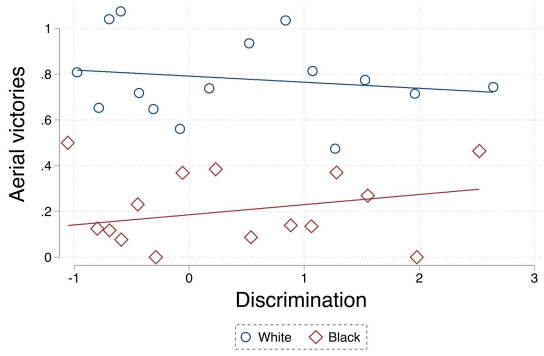
Figure B.7: Schooling and Aerial Victories vs. Discrimination of Residence County in the 1990s Index to Public Records




(A) Schooling in 1940



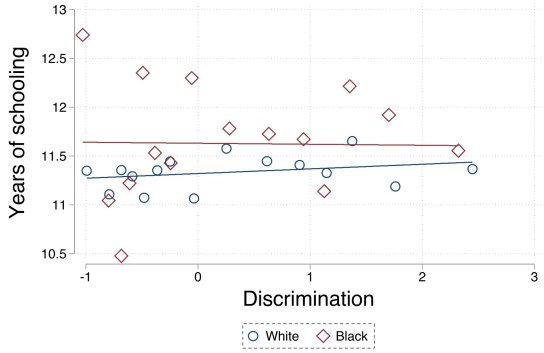

(B) Schooling at enlistment



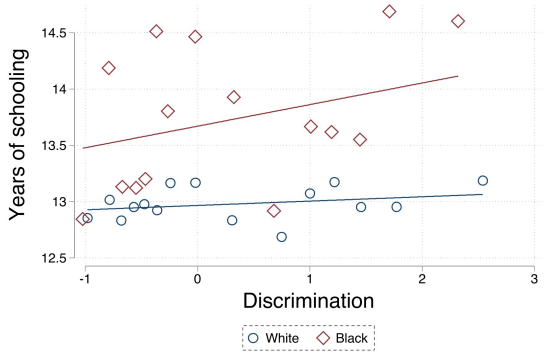

(C) Aerial victories

Notes: Binned scatter plots of years of education in 1940, years of education at enlistment, and raw aerial victories against the racial discrimination index of the residence county in the 1990s Index to Public Records, separately for Black and white pilots. Schooling measures are residualized by year of birth; aerial victories are raw wartime credit counts. Each dot represents the mean within an equal-sized bin; up to 15 bins are used. The lines show the linear fit.

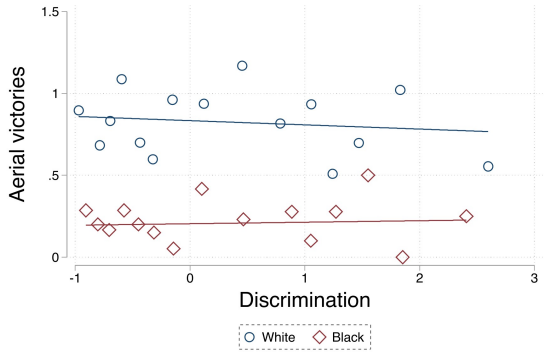
Figure B.8: Schooling and Aerial Victories vs. Discrimination of Residence County in the NUMIDENT




(A) Schooling in 1940




(B) Schooling at enlistment




(C) Aerial victories

Notes: Binned scatter plots of years of education in 1940, years of education at enlistment, and raw aerial victories against the racial discrimination index of the residence county in the NUMIDENT, separately for Black and white pilots. Schooling measures are residualized by year of birth; aerial victories are raw wartime credit counts. Each dot represents the mean within an equal-sized bin; up to 15 bins are used. The lines show the linear fit.

Table B.10: The Effect of Racial Discrimination on Black Pilots' Outcomes (1950 Census and 1960–90 City Directories) with CEM

| | 1950 Census | | | | | | 1960-90 City Directories |
|---|---|---|---|---|---|---|---|
| | Income | Prestige | Cognitive | Leadership | Social | Z-score | Z-score |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| Black × Discrimination | -774.431*** | -9.011*** | -0.646*** | -0.084 | -0.730*** | -0.694*** | -0.093 |
| | (124.988) | (2.069) | (0.132) | (0.139) | (0.101) | (0.111) | (0.100) |
| Black | 470.818*** | 7.841** | 0.324 | -0.043 | -0.175 | 0.395** | -0.416*** |
| | (163.115) | (3.090) | (0.227) | (0.349) | (0.228) | (0.176) | (0.112) |
| Discrimination | -283.387*** | -1.268 | -0.298** | -0.200* | -0.031 | -0.213** | 0.019 |
| | (78.279) | (2.128) | (0.140) | (0.115) | (0.132) | (0.091) | (0.117) |
| dydx Margin on Black | -275.216** | -0.839 | -0.298** | -0.124 | -0.878*** | -0.274*** | -0.476*** |
| | (138.059) | (2.082) | (0.143) | (0.352) | (0.255) | (0.095) | (0.123) |
| Adj. $R^2$ | 0.286 | 0.219 | 0.237 | 0.103 | 0.340 | 0.260 | 0.183 |
| $N$ | 356 | 356 | 356 | 356 | 356 | 356 | 352 |

Notes: Outcomes are occupation-based: median income in 1950 dollars, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. This table shows regression results using outcomes from the 1950 Census and 1960–90 City Directories for a sample of Black and white pilots with the same residence county, similar education level at the time of enlistment (high school or college), similar aerial victories (by median) and pilot type with Coarsened Exact Matching. Standard errors (in parentheses) are clustered at the residence state level. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table B.11: The Effect of Racial Discrimination on Black Pilots' Outcomes with CEM (1990s Index to Public Records and NUMIDENT)

| | House percentile (1) | ZIP code income percentile (2) |
|---|---|---|
| Black × Discrimination | -4.959*** | -9.371*** |
| | (1.359) | (2.690) |
| Black | -4.842 | -12.936*** |
| | (2.894) | (3.629) |
| Discrimination | 3.398* | 2.168 |
| | (1.758) | (3.137) |
| dydx Margin on Black | -8.576*** | -19.260*** |
| | (3.032) | (3.636) |
| Adj. $R^2$ | 0.568 | 0.331 |
| $N$ | 475 | 340 |

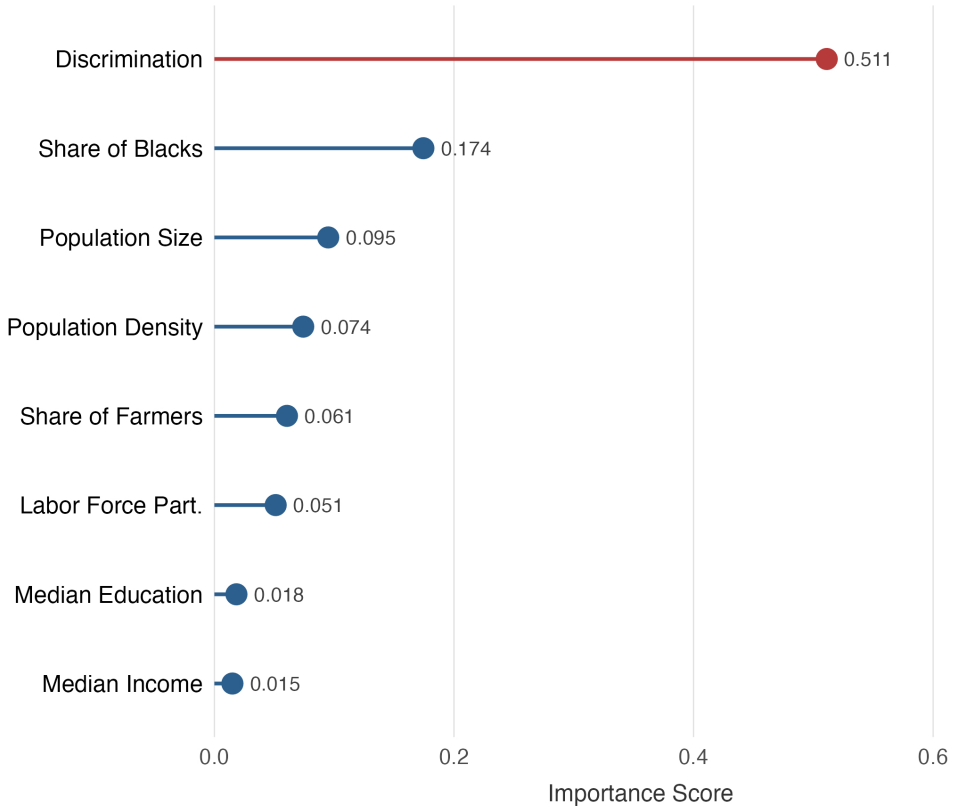
Notes: This table shows regression results using pilot outcomes from the 1990s Index to Public Records and NUMIDENT. We regress each outcome on a Black indicator, a continuous measure of racial discrimination in the residence county when the outcome was measured, and their interaction, for a sample of Black and white pilots with the same residence county, similar education level at the time of enlistment (high school or college), similar aerial victories (by median), and pilot type with Coarsened Exact Matching. Standard errors (in parentheses) are clustered at the residence state level. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Table B.12: High- versus Low-Discrimination CEM Within States

| | 1950 Census | | | | | | 1960–90 City Directories | 1990s IPR | NUMIDENT |
|---|---|---|---|---|---|---|---|---|---|
| | Income | Prestige | Cognitive | Leadership | Social | Z-score | Z-score | House percentile | ZIP income percentile |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) |
| Black × High-discrimination county | -1165.870*** | -6.199* | -1.139*** | -0.664 | -1.434** | -0.923** | -0.648*** | -6.829 | -19.048*** |
| | (226.157) | (3.152) | (0.376) | (0.781) | (0.631) | (0.370) | (0.180) | (4.810) | (5.997) |
| Adj. $R^2$ | 0.056 | 0.064 | 0.036 | 0.059 | 0.145 | 0.024 | 0.115 | 0.402 | 0.133 |
| Matched *N* | 1,114 | 911 | 1,106 | 905 | 905 | 1,115 | 1,128 | 1,391 | 1,122 |
| High-discrimination *N* | 708 | 581 | 698 | 581 | 581 | 709 | 557 | 681 | 556 |
| Low-discrimination *N* | 406 | 330 | 408 | 324 | 324 | 406 | 571 | 710 | 566 |
| States | 40 | 36 | 40 | 36 | 36 | 40 | 36 | 40 | 36 |

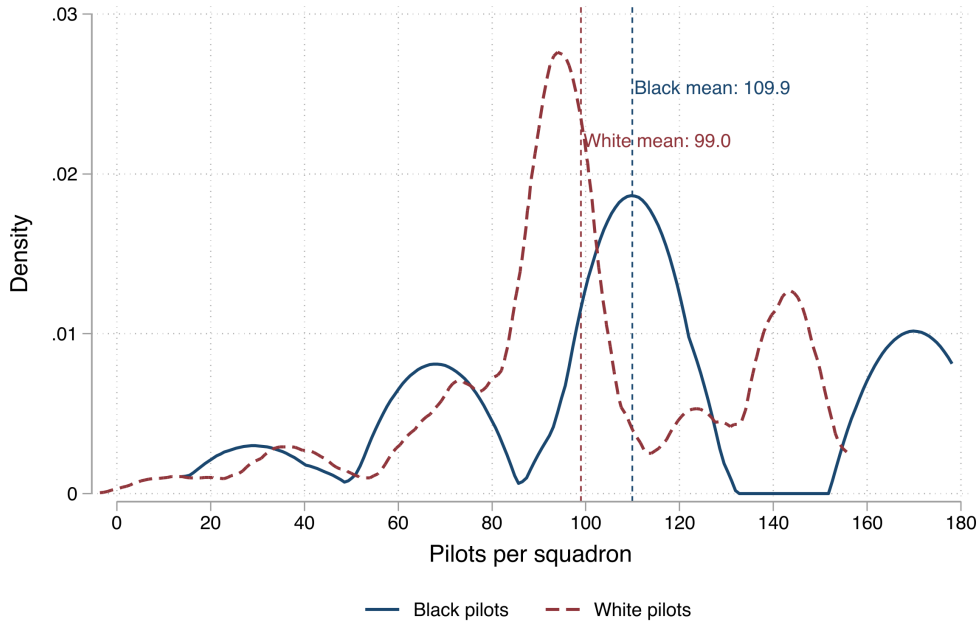
Notes: The first seven columns report occupation-based outcomes: median income in 1950 dollars, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and composite occupational z-scores. This table reports pooled regressions using CEM weights. High-discrimination county equals one if the county discrimination index is above the mean among pilots in the same state for the relevant outcome source. CEM exact-matches on state, race, and pilot type, and on median bins of the 1940 county occupational income score, urban share, manufacturing employment share, and high-school graduation share. The regression includes the original 1940 county characteristics and their interactions with Black. Standard errors (in parentheses) are clustered at the residence state level. * $p<0.1$, ** $p<0.05$, *** $p<0.01$

Figure B.9: Variable Importance for Black-White Occ *z-score* Gap in 1950




Notes: This figure shows the variable importance measures derived from a causal forest algorithm used to estimate the heterogeneous effects of the Black indicator on 1950 occ z-score (Athey et al., 2019). We use a double residualization approach, orthogonalizing both the outcome and the Black indicator with respect to residence state fixed effects. The figure reports the importance score for each county-level covariate, calculated as the weighted frequency with which a variable is used to split nodes across the forest's trees. The covariates include the discrimination index, population size, population density, the share of Blacks, labor force participation, median education, median income, and the share of farmers. Higher values indicate that the variable contributes more to explaining effect heterogeneity.

Figure B.10: Squadron-Size Densities by Race in the Peer-Origin Construction Sample




Notes: This figure plots kernel densities of squadron size separately for Black and white pilots in the sample used to construct leave-one-out squadron peer origins. Squadron size is measured as the number of unique pilots in a pilot's AAF squadron among pilots with nonmissing squadron assignment and birth county. Dashed vertical lines mark race-specific means.

## Table B.13: Permutation Test for 1940 Birth-County Characteristics

| Birth-county characteristic | Obs. $\hat{\beta}$ | Obs. $\lvert t\rvert$ | Perm. mean $\lvert t\rvert$ | Permutation $p$-value | $N$ |
|---|---|---|---|---|---|
| *Demographic* | | | | | |
| Black share | -0.098 | 2.21 | 1.43 | 0.210 | 4,070 |
| Foreign-born share | -0.003 | 0.19 | 1.37 | 0.914 | 4,075 |
| Female share | 0.043 | 1.64 | 1.27 | 0.299 | 4,075 |
| Population share age 15–24 | 0.015 | 0.85 | 1.37 | 0.601 | 4,072 |
| Population share age 65+ | 0.009 | 0.32 | 1.29 | 0.839 | 4,072 |
| *Education* | | | | | |
| High-school graduate share | 0.035 | 2.13 | 1.33 | 0.196 | 4,070 |
| School attendance share, age 7–13 | -0.011 | 0.83 | 1.36 | 0.630 | 4,075 |
| School attendance share, age 14–15 | 0.010 | 0.61 | 1.37 | 0.713 | 4,075 |
| School attendance share, age 16–17 | 0.029 | 1.38 | 1.33 | 0.392 | 4,075 |
| *Labor and local economy* | | | | | |
| Occupational income score | 0.042 | 2.65 | 1.30 | 0.110 | 4,070 |
| Male labor-force participation | -0.005 | 0.16 | 1.27 | 0.911 | 4,075 |
| Female labor-force participation | 0.046 | 1.46 | 1.36 | 0.383 | 4,075 |
| Professional employment share | 0.034 | 2.08 | 1.34 | 0.219 | 4,075 |
| Manufacturing establishments per 1,000 people | 0.005 | 0.28 | 1.43 | 0.875 | 4,049 |
| Retail stores per 1,000 people | 0.031 | 1.31 | 1.35 | 0.442 | 4,072 |
| Wholesale establishments per 1,000 people | 0.016 | 1.00 | 1.40 | 0.563 | 4,072 |
| *Housing and amenities* | | | | | |
| Owner-occupied dwelling share | -0.052 | 1.60 | 1.28 | 0.317 | 4,075 |
| Good-condition dwelling share | 0.021 | 1.24 | 1.29 | 0.438 | 4,074 |
| Electric-light dwelling share | 0.021 | 1.30 | 1.32 | 0.441 | 4,074 |
| Radio dwelling share | 0.005 | 0.27 | 1.39 | 0.864 | 4,074 |
| *Agriculture* | | | | | |
| Tenant-farm share | -0.006 | 0.29 | 1.30 | 0.855 | 3,979 |
| Share-tenant/cropper farm share | -0.014 | 0.82 | 1.38 | 0.623 | 3,976 |
| Average farm size | -0.009 | 0.52 | 1.35 | 0.763 | 4,037 |
| Farm value per farm acre | 0.001 | 0.05 | 1.28 | 0.981 | 3,979 |
| *Labor-market distress* | | | | | |
| Registered unemployed share, 1937 | 0.005 | 0.24 | 1.31 | 0.886 | 4,070 |
| Emergency-worker share, 1937 | 0.004 | 0.27 | 1.27 | 0.865 | 4,070 |
| Partly unemployed share, 1937 | 0.029 | 1.42 | 1.34 | 0.395 | 4,070 |

Notes: This table tests whether squadron peer composition is unusually sorted on predetermined birth-county characteristics. Characteristics are grouped by domain. For each characteristic, we compute the leave-one-out peer mean within the pilot's actual squadron, standardize the pilot and peer variables, and regress the pilot's own characteristic on the peer mean with race $\times$ pilot type fixed effects and robust standard errors. We then permute squadron IDs within race $\times$ pilot type cells 1,000 times and recompute the statistic. The permutation $p$-value is the share of placebo assignments with an absolute $t$-statistic at least as large as the observed statistic, using a one-draw correction. Most county characteristics come from the ICPSR 2896 county aggregates for 1940 (DS0032 and DS0033); occupational income score, Black share, and high-school graduate share come from project aggregates of the person-level 1940 Census. The sample excludes pilots missing squadron assignment, birth county, race, pilot type, or the relevant birth-county characteristic.

Table B.14: Balance Test of Peer-Origin Instrument

| Variable | $Z_i$ | $Black_i \times Z_i$ | $N$ |
|---|---|---|---|
| Education, 1940 Census | -0.14 | 0.04 | 1,248 |
| | (0.10) | (0.15) | |
| Education at enlistment | -0.20 | 0.04 | 920 |
| | (0.09) | (0.15) | |
| Aerial victories | 0.04 | -0.12 | 1,745 |
| | (0.08) | (0.11) | |

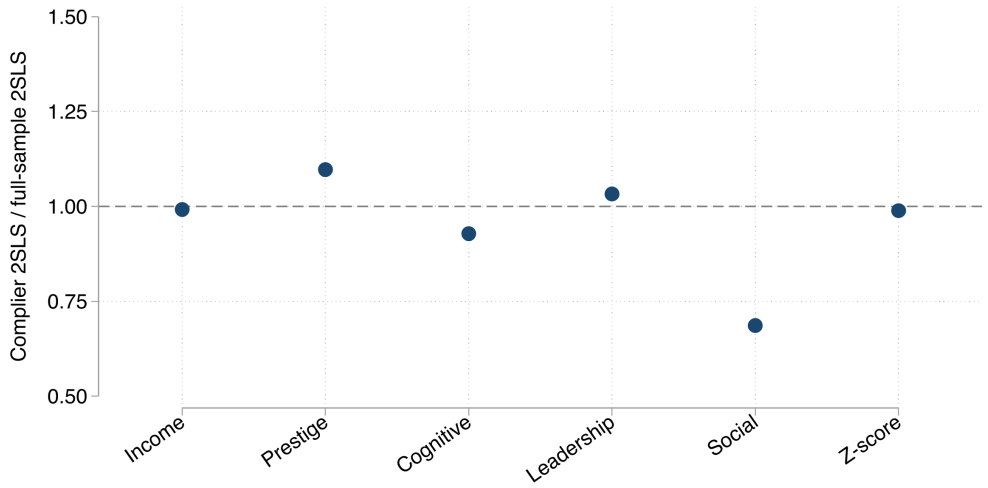
Notes: Each row reports a separate regression of the listed pilot characteristic on the peer-origin instrument $Z_i$, its interaction with Black status, and a Black indicator. The table reports the coefficients on $Z_i$ and $Black_i \times Z_i$; the Black coefficient is included in the regression but not reported. All specifications absorb residence-state, birth-state, pilot-type, and birth-year fixed effects. Standard errors clustered by 1950 residence state are reported in parentheses. Aerial victories are credited wartime victories and should be interpreted as a service-performance diagnostic rather than a predetermined characteristic.

Table B.15: Zero-Stage Analysis: Peer Origins and 1950 Residence

| | Indicator of pilot $i$ living in county $c$ | | |
|---|---|---|---|
| | (1) | (2) | (3) |
| Peer-origin share ($s_{i,c}$) | 0.25*** | 0.25*** | 0.03*** |
| | (0.01) | (0.01) | (0.01) |
| Distance kernel ($K(d_{b_i,c})$) | 0.02*** | 0.02*** | 0.02*** |
| | (0.00) | (0.00) | (0.00) |
| Adj. $R^2$ | 0.008 | 0.009 | 0.015 |
| $N$ | 8,337,761 | 8,337,761 | 8,337,761 |
| Pilot FE | No | Yes | Yes |
| Destination county FE | No | No | Yes |

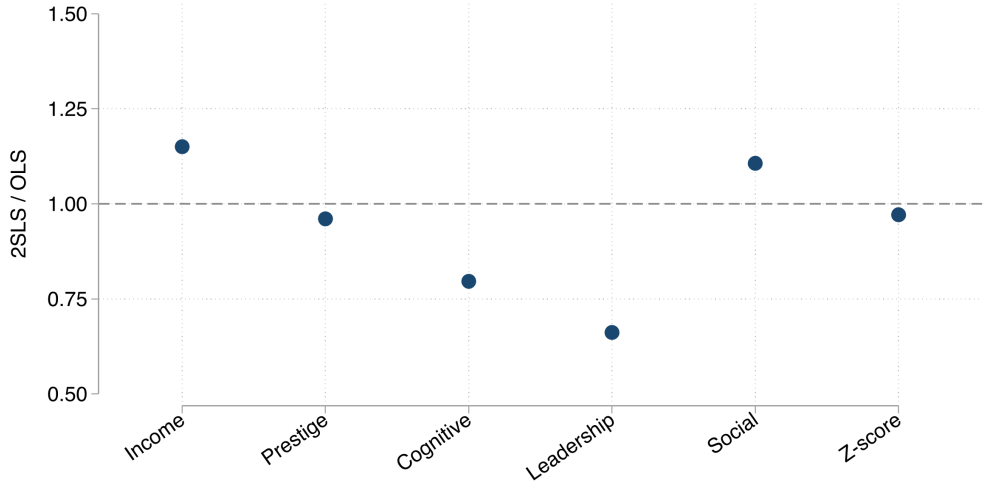
Notes: This table reports the zero-stage analysis linking a pilot's residence county in 1950 to the leave-one-out share of squadron peers born in the county. The data are constructed at the pilot-county pair level, where the dependent variable is an indicator equal to one if the pilot lived in county $c$ in 1950 and zero otherwise. The specification controls for the same inverse-distance kernel $K(d)$ used in the IV construction, with distance measured in 100-kilometer units. Standard errors are clustered at the pilot level and reported in parentheses. * $p$<0.1, ** $p$<0.05, *** $p$<0.01

Figure B.11: Ratio of 2SLS Estimates for Compliers and All Pilots



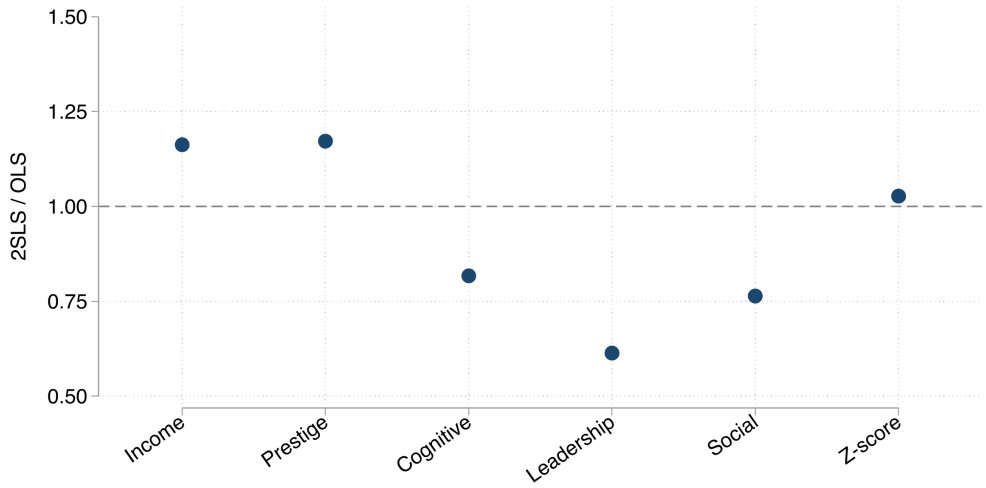

Notes: Outcome labels refer to occupational median income, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. Ratios compare complier and full-sample 2SLS coefficients on Black × discrimination for 1950 outcomes. Compliers' 1950 residence counties are among their five most likely destinations. Predictions use the leave-one-out share of squadron peers born in each county and the inverse-distance kernel from the pilot's birth county. Compliers account for 41.7% of the full IV analysis sample (858 of 2056 pilots); the shares are 42.5% among white pilots and 37.0% among Black pilots. The dashed line denotes equal estimates.

Figure B.12: 2SLS-to-OLS Ratios for Black × Discrimination Estimates




Notes: Outcome labels refer to occupational median income, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. Ratios compare 2SLS and OLS coefficients on Black × discrimination for 1950 occupational outcomes. The dashed line denotes equal estimates.

Figure B.13: 2SLS-to-OLS Ratios for Compliers




Notes: Outcome labels refer to occupational median income, prestige, cognitive skill requirements, leadership skill requirements, social skill requirements, and a composite occupational z-score. This figure reports the ratio of the 2SLS estimate to the corresponding OLS estimate for the Black $\times$ discrimination coefficient across 1950 occupational outcomes, restricting the sample to compliers. Compliers' 1950 residence counties are among their five most likely destinations. Predictions use the leave-one-out share of squadron peers born in each county and the inverse-distance kernel from the pilot's birth county. Compliers account for 41.7% of the full IV analysis sample (858 of 2056 pilots); the shares are 42.5% among white pilots and 37.0% among Black pilots. The dashed horizontal line marks equality between 2SLS and OLS.

Table B.16: Robustness Checks Controlling for Peer-Origin Socioeconomic Status

| | Income | Prestige | Cognitive | Leadership | Social | Z-score |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: 2SLS* | | | | | | |
| Black × Discrimination | -1040.57*** | -12.32*** | -0.78*** | -0.14 | -0.78*** | -0.89*** |
| | (199.32) | (2.58) | (0.18) | (0.28) | (0.20) | (0.14) |
| Black | -2030.84 | 5.98 | -0.51 | -3.92*** | -2.29 | -1.03 |
| | (1265.94) | (17.95) | (1.53) | (1.49) | (1.54) | (1.00) |
| Discrimination | 339.51 | 4.13 | 0.53 | 0.60 | 0.36 | 0.44 |
| | (433.47) | (4.42) | (0.41) | (0.45) | (0.37) | (0.34) |
| Peer-Origin SES Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| *N* | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 |
| *Panel B: Reduced Form* | | | | | | |
| Black × Z | -466.37*** | -5.52*** | -0.34*** | -0.04 | -0.34*** | -0.39*** |
| | (89.92) | (1.89) | (0.09) | (0.12) | (0.10) | (0.10) |
| Black | 278.67 | 33.11 | 0.70 | -4.63** | -0.75 | 0.68 |
| | (1614.87) | (26.37) | (1.88) | (2.07) | (1.94) | (1.61) |
| Z | 111.67* | 1.34* | 0.12** | 0.08 | 0.10* | 0.11** |
| | (62.86) | (0.75) | (0.05) | (0.05) | (0.05) | (0.05) |
| Peer-Origin SES Controls | Yes | Yes | Yes | Yes | Yes | Yes |
| *N* | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 | 2,056 |
| Adj. $R^2$ | 0.046 | 0.021 | 0.036 | 0.051 | 0.100 | 0.031 |

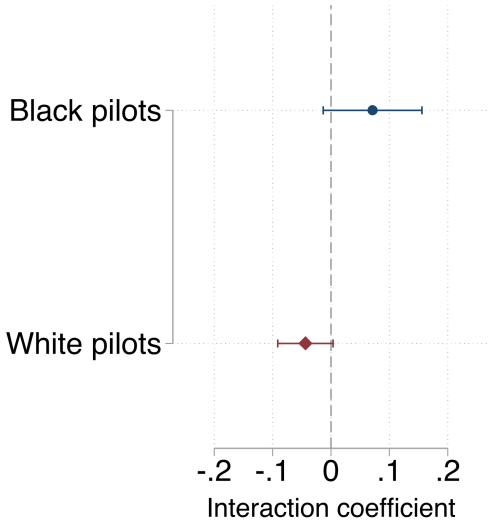
Notes: Relative to Table 5, specifications add the weighted 1940 peer-origin occupational score for each pilot's race and its interaction with Black. All other details follow Table 5.

Table B.17: 2SLS and Reduced Form Results with Race-Specific Fixed Effects

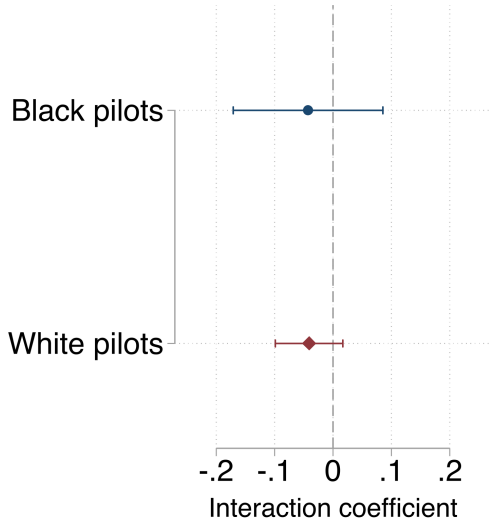
| | Income | Prestige | Cognitive | Leadership | Social | Z-score |
|---|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: 2SLS* | | | | | | |
| Black × Discrimination | -914.55** | -1.60 | -1.19** | -1.16 | -1.73** | -0.90** |
| | (394.09) | (7.97) | (0.51) | (0.96) | (0.87) | (0.45) |
| Discrimination | 144.15 | 1.80 | 0.16 | 0.01 | 0.07 | 0.14 |
| | (361.33) | (3.70) | (0.31) | (0.34) | (0.26) | (0.28) |
| Black × Predetermined FE | Yes | Yes | Yes | Yes | Yes | Yes |
| S-W *F*: Discrimination | 19.82 | 19.82 | 19.82 | 19.82 | 19.82 | 19.82 |
| S-W *F*: Black × Discrimination | 11.35 | 11.35 | 11.35 | 11.35 | 11.35 | 11.35 |
| *N* | 2,050 | 2,050 | 2,050 | 2,050 | 2,050 | 2,050 |
| *Panel B: Reduced Form* | | | | | | |
| Black × Z | -195.18* | -0.24 | -0.26** | -0.26 | -0.38* | -0.19* |
| | (117.28) | (1.73) | (0.11) | (0.20) | (0.20) | (0.10) |
| Z | 29.06 | 0.31 | 0.03 | 0.01 | 0.02 | 0.03 |
| | (62.42) | (0.64) | (0.05) | (0.06) | (0.05) | (0.05) |
| Black × Predetermined FE | Yes | Yes | Yes | Yes | Yes | Yes |
| *N* | 2,050 | 2,050 | 2,050 | 2,050 | 2,050 | 2,050 |
| Adj. $R^2$ | 0.041 | 0.035 | 0.032 | 0.067 | 0.107 | 0.033 |

Notes: Relative to Table 5, specifications interact the birth-state, birth-year, and pilot-type fixed effects with Black in both stages. S-W *F* is the Sanderson–Windmeijer first-stage *F*-statistic. All other details follow Table 5.

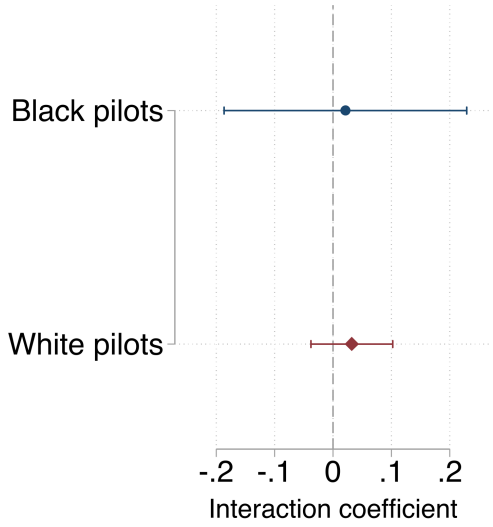
Figure B.14: First-Stage Heterogeneity by Ability Proxy




(A) Schooling in 1940



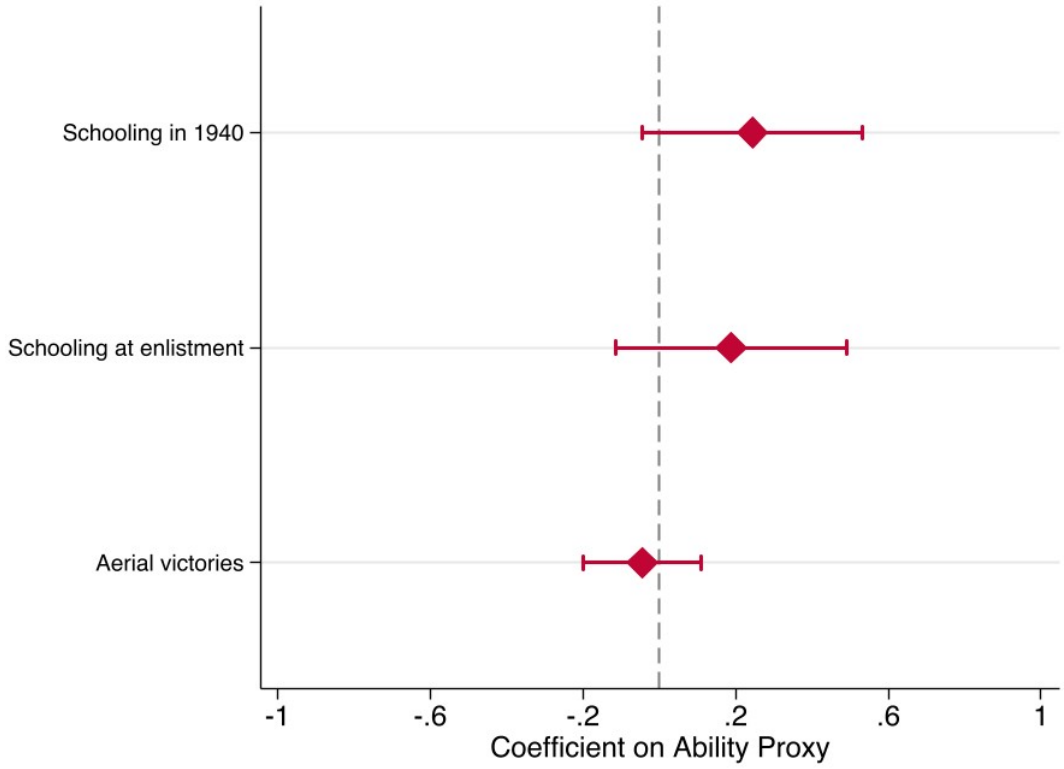

(B) Schooling at Enlistment




(C) Aerial Victories

Notes: Points report the coefficients on the interaction between the peer-origin instrument and each continuous ability proxy; bars show 95% confidence intervals. For each proxy, we regress the discrimination index of the pilot's 1950 residence county on the instrument, the ability proxy, and their interaction, separately for Black and white pilots. All specifications absorb residence-state, birth-state, birth-year, and pilot-type fixed effects.

Figure B.15: Relationship between Ability Proxies and Change in Discrimination, Black Pilot Movers




Notes: Each row reports the coefficient from a separate OLS regression of the change in county-level discrimination (destination minus origin) on the indicated ability proxy, estimated on Black pilot movers. Proxies are standardized to mean zero and unit variance. Horizontal bars show 95 percent confidence intervals based on standard errors clustered at the origin county. Schooling in 1940 is measured from the 1940 Census; schooling at enlistment is measured from military enlistment records; aerial victories are measured as the number of enemy aircraft credited during wartime service.

Table B.18: Balance Table: Enlistees With vs. Without AGCT Scores

| | (1) Mean Has AGCT | (2) Mean No AGCT | (3) Standardized Difference | (4) *p*-value Equiv. Test |
|---|---|---|---|---|
| Southern enlistment state | 0.263 | 0.294 | -0.069 | 0.000 |
| Enlistment year | 1943.0 | 1942.7 | 0.168 | 1.000 |
| Birth year | 1919.7 | 1918.1 | 0.252 | 1.000 |
| Education (years) | 10.6 | 10.5 | 0.042 | 0.000 |
| US citizen | 0.982 | 0.990 | -0.079 | 0.000 |
| Foreign born | 0.040 | 0.030 | 0.056 | 0.000 |
| Married | 0.265 | 0.231 | 0.080 | 0.000 |
| *N* | 442,282 | 7,759,667 | | |

Notes: This table compares means of key covariates between WWII Army enlistees with and without AGCT scores. The sample includes Black and white male enlistees with non-missing birth year and birthplace state. Southern enlistment state is defined using the Census South region (South Atlantic, East South Central, and West South Central divisions). Education is converted to approximate years of schooling. Column (3) reports the standardized difference (difference in means divided by the pooled standard deviation). Column (4) reports the *p*-value from the two one-sided *t*-tests (TOST) procedure with a 0.1 standard deviation margin; a *p*-value below 0.05 indicates that the two groups are statistically equivalent within 0.1 SD.

Table B.19: Balance Table: Census-Matched vs. Unmatched Enlistees, Conditional on AGCT Availability

| | (1) Mean Census-matched | (2) Mean Unmatched | (3) Standardized Difference | (4) *p*-value Equiv. Test |
|---|---|---|---|---|
| Southern enlistment state | 0.249 | 0.277 | -0.062 | 0.000 |
| Birth year | 1919.6 | 1919.8 | -0.028 | 0.000 |
| Education (years) | 10.7 | 10.5 | 0.093 | 0.010 |
| US citizen | 0.984 | 0.979 | 0.040 | 0.000 |
| Foreign born | 0.036 | 0.044 | -0.043 | 0.000 |
| Married | 0.281 | 0.248 | 0.076 | 0.000 |
| AGCT score | 99.2 | 96.2 | 0.127 | 1.000 |
| *N* | 225,023 | 217,259 | | |

Notes: This table compares means of key covariates between enlistees successfully linked to the 1950 Census and those not linked, conditional on having an AGCT score. Enlistment year is excluded because it is constant (1943) in this subsample. Southern enlistment state is defined using the Census South region. Education is converted to approximate years of schooling. Column (3) reports the standardized difference (difference in means divided by the pooled standard deviation). Column (4) reports the *p*-value from a TOST equivalence test with a 0.1 standard deviation margin; a *p*-value below 0.05 indicates that the two groups are statistically equivalent within 0.1 SD.

Table B.20: Balance Table: NUMIDENT-Matched vs. Unmatched Enlistees, Conditional on AGCT Availability

| | (1) Mean SSDI-matched | (2) Mean Unmatched | (3) Standardized Difference | (4) *p*-value Equiv. Test |
|---|---|---|---|---|
| Southern enlistment state | 0.249 | 0.269 | -0.046 | 0.000 |
| Birth year | 1921.0 | 1919.2 | 0.343 | 1.000 |
| Education (years) | 10.7 | 10.6 | 0.080 | 0.000 |
| US citizen | 1.000 | 0.974 | 0.193 | 1.000 |
| Foreign born | 0.000 | 0.057 | -0.291 | 1.000 |
| Married | 0.222 | 0.283 | -0.139 | 1.000 |
| AGCT score | 98.9 | 97.2 | 0.074 | 0.000 |
| *N* | 133,130 | 309,152 | | |

Notes: This table compares means of key covariates between enlistees successfully linked to the Social Security Death Index (SSDI) and those not linked, conditional on having an AGCT score. Enlistment year is excluded because it is constant (1943) in this subsample. Southern enlistment state is defined using the Census South region. Education is converted to approximate years of schooling. Column (3) reports the standardized difference (difference in means divided by the pooled standard deviation). Column (4) reports the *p*-value from a TOST equivalence test with a 0.1 standard deviation margin; a *p*-value below 0.05 indicates that the two groups are statistically equivalent within 0.1 SD.

Table B.21: Summary Statistics for Enlistees (Public Data)

| | Black | | | | White | | | |
|---|---|---|---|---|---|---|---|---|
| | Mean | SD | Median | N | Mean | SD | Median | N |
| Birth year | 1,920 | 5 | 1,922 | 46,528 | 1,920 | 6 | 1,923 | 395,754 |
| AGCT | 70 | 20 | 67 | 46,528 | 101 | 22 | 103 | 395,754 |
| Years of schooling (Enlistment Records) | 9.47 | 1.87 | 8.00 | 46,528 | 10.74 | 2.15 | 11.00 | 395,754 |
| Income ($) 1950 | 2,109 | 1,876 | 1,850 | 3,647 | 2,908 | 2,050 | 2,750 | 38,530 |
| Occ income score ($) 1950 | 2,170 | 728 | 2,000 | 15,038 | 2,809 | 965 | 2,500 | 172,363 |
| Occ prestige score 1950 | 28.14 | 10.90 | 28.40 | 15,038 | 37.08 | 12.22 | 35.50 | 172,363 |
| Occ cognitive score 1950 | -0.41 | 0.78 | -0.69 | 15,038 | 0.05 | 1.01 | -0.26 | 172,363 |
| Occ leadership score 1950 | -0.28 | 0.76 | -0.42 | 15,038 | 0.03 | 1.01 | -0.15 | 172,363 |
| Occ social skill score 1950 | -0.50 | 0.79 | -0.87 | 15,038 | 0.04 | 1.01 | -0.07 | 172,363 |
| SES z-score 1950 | -0.58 | 0.78 | -0.80 | 15,038 | 0.07 | 1.01 | -0.20 | 172,363 |
| Discrimination 1950 | 1.24 | 0.92 | 1.27 | 15,038 | 0.26 | 0.92 | 0.05 | 172,363 |
| ZIP code income percentile death | 36.83 | 32.10 | 27.00 | 11,568 | 53.77 | 30.01 | 55.00 | 116,008 |
| Discrimination death | 1.04 | 0.97 | 1.08 | 11,568 | 0.24 | 0.94 | -0.03 | 116,008 |